When Does Crime Enter House Prices? Evidence from Chicago

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Abstract

I estimate how crime affects housing prices in Chicago. Although homebuyers presumably care about crime, most offenses have a near-zero effect on sale prices or rents. I argue that this is consistent with limited information: crime only changes buyers' valuations if they learn of an incident and revise their expectations about future exposure. First, price effects are hyperlocal even though the predictive value of crimes is not. Second, price effects strengthen with agents' local experience. Third, even within an offense, the effect depends on where at the property the incident occurred. These patterns are consistent with buyers responding to incidents they learn about rather than to the crime those incidents forecast.

1 Introduction

A home is most homeowners' largest asset and an important consumption good. Neighborhood crime threatens both dimensions: it undermines the safety of the environment in which families live and the value of the asset they hold. In a 2025 Redfin survey, personal safety and a low crime rate were the two most cited non-negotiables among prospective homebuyers.[1]

This paper asks when crime affects housing prices and why. Which offenses are capitalized, and what distinguishes them from those that are not? Where does the discount emerge: in the price a seller asks, or in what a buyer ultimately pays? And which offenses lower rents?

I answer these questions using data from Chicago. I combine 530,395 single-family, condominium, and residential-income transactions on 337,461 properties from the Chicago Multiple Listing Service, spanning 2007 to 2025, with 5.59 million geocoded crime incidents, of which I analyze twenty-two offense categories. The Chicago Police Department recorded 5,657,404 offenses over this period, 5,591,677 of them (98.8 percent) geocoded. The transaction data follow properties

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[1]A 2025 Redfin/Ipsos survey of 1,224 prospective homebuyers finds that personal safety (78%) and a low crime rate (74%) are the two most frequently cited "must-have" attributes among eighteen home and neighborhood features (Anderson, 2025).

from listing through sale, allowing me to distinguish the initial asking price from the final sale price and to observe time on market, withdrawals, and returns to market. Rental listings provide the corresponding outcomes in the lease market. The crime data give coordinates for each incident, allowing exposure to be measured around each property.

My identification strategy uses a repeat-sales framework, comparing the same parcel across multiple transactions, adjusting for tract-by-month fixed effects, and further differencing against crime in a 0.20–0.30 mile annulus around the same parcel (finer than the tract). Parcel fixed effects absorb permanent differences across properties; tract-by-month fixed effects absorb contemporaneous shocks shared by the surrounding neighborhood; and the near–far contrast allows for further within-tract adjustments. Exposure within 0.05 miles (264 feet) is contrasted with exposure in a 0.20–0.30 mile annulus; estimates across nine near–far ring definitions are reported in Appendix Table A2. Nonparametric distance profiles motivate these bands: for the property offenses, the price response is concentrated near the property and indistinguishable from zero beyond roughly a block. Moving the treatment band off the doorstep tests where the effect ends: the property-crime discounts disappear beyond the block, while homicide's extends a few blocks (Section 5.6). I run this design for twenty-two offense categories on sale and lease listings alike, and validate each estimate using a placebo on crime occurring after the transaction, which cannot affect a price already set.

I find that some crimes affect housing prices, while most have near-zero effects. Of the twenty-two reported offense categories, four (homicide, burglary, narcotics, and vandalism) carry a sale-price discount across multiple exposure windows that passes the post-sale placebo at the offense's central window (burglary's placebo is significant at two and six months; two more, disorderly conduct and sex offenses, are statistically significant at a single long-horizon window). Homicide, among the rarest offenses, carries the largest discount at about three percent per incident, while common or heavily feared offenses such as theft, motor-vehicle theft, assault, and robbery show no detected discount. An offense can therefore be severe, frequent, or widely feared and still show no detectable discount.

Why, then, are effects confined to particular offenses, when buyers are presumably averse to crime in general? A simple framework helps us think about this question. For an incident to affect the house price, three conditions must hold: (a) buyers must learn of the incident; (b) learning of it must raise their expectation of future crime of that type at the property; and (c) buyers must be willing to pay to avert that crime at the property. The first condition concerns whether information arrives, the second what it implies about future crime, and the third what avoiding that crime is worth: a buyer who sees a car stolen from the curb may expect more car thefts on the block and still pay no less for the house.

Three lines of evidence help explain this pattern. The first concerns observability. Holding the offense fixed, I compare incidents at a dwelling's frontage with those inside it. For burglary, incidents at the frontage are associated with discounts in seven of twelve windows and incidents inside the dwelling in none, and the difference between the two grows with the exposure window

and is significant at twelve months; vandalism shows the opposite pattern (Section 5.2). Agent experience results point to how that information travels. For homicide, discounts are larger when the buyer's agent has more experience in the same neighborhood, also after controlling for the agent's total experience (Table OA25). Although asking prices do not respond to crime on average, they fall with some nearby crimes when the listing agent is more experienced; for narcotics, this gradient reflects the agent's total rather than local experience. Agents are not randomly assigned to transactions, so these patterns are associations rather than causal effects. The second concerns relevance. The four priced offenses share a risk that stays with the address: to the house (burglary, vandalism), to the people living in it (homicide), or to the block in front of it (narcotics), while motor-vehicle theft, whose loss falls on a movable, insured asset, is a precise zero. This pattern is consistent with relevance but does not test it, since offenses differ in many ways at once. The closer test holds the offense fixed: within vandalism, damage to property lowers sale prices in half of the exposure windows and damage to vehicles in none, although neither estimate is significant at the central window (Section 5.2). The third concerns how buyers use crime information once they observe it. Crime in the surrounding neighborhood, a fifth to three tenths of a mile from the property, forecasts future crime near the property for nearly every offense, priced and unpriced alike. Yet discounts do not rise with how strongly an offense predicts its own recurrence. Narcotics, which recurs most predictably, carries a small discount, while homicide, which does not recur, carries the largest; non-domestic assault recurs four times more predictably than burglary and carries no detectable discount. Recurrence alone does not determine which offenses are capitalized.

The second set of results shows that sale-price discounts generally emerge after listing rather than in the seller's initial ask. Asking prices adjust little overall, and the discount appears at the close; narcotics also lowers the sale-to-list ratio, and a nearby homicide lengthens time on market by roughly six days against a median of 42 and raises the probability of withdrawal. The broader pattern is that crime is priced in negotiation rather than in the posted price.

Finally, I apply the same design to rental listings. Burglary lowers asking rents under parcel fixed effects, and sex offenses and liquor violations lower them under parcel-by-era but not parcel fixed effects. Narcotics leaves asking rents unchanged but raises the probability that a rental listing returns to market, and homicide's rental estimate is too imprecise to sign. Because the data record asking rents rather than negotiated rents, these estimates describe what landlords post.

The estimates are robust to time-varying block characteristics and the choice of distance bands, and inference is robust to Conley spatial standard errors. Adding building permits, vacant buildings, business licenses, tree canopy, and total 311 service requests, each measured as the near–far stock twelve months before sale, moves the coefficients by hundredths of a percentage point, although with all five controls burglary and narcotics are significant only at ten percent, so the discount does not appear to reflect these measured block characteristics correlated with local crime. Conley standard errors at 0.5 and 1.0 km, reported in Table OA32, are below the clustered standard errors for every priced offense; inference uses the clustered ones (Conley, 1999). With 77 community-area clusters, homicide and burglary remain significant at five percent and vandalism at ten percent,

while narcotics is not ($p = 0.12$; Table OA32). For the property offenses, the discount is concentrated in the innermost ring across nine near–far ring definitions (Appendix Table A2), so the contrast is not an artifact of the chosen cutoffs.

This paper makes three contributions.

First, I measure crime exposure at the property and incident level across a broad set of offenses and use this common design to examine why some offenses are capitalized while others are not. Studies using local crime rates typically assign exposure at more aggregate geographies (Thaler, 1978; Schwartz et al., 2003; Gibbons, 2004; Ihlanfeldt and Mayock, 2010), making estimates sensitive to the spatial units used to construct exposure, the modifiable areal unit problem (MAUP), and making it difficult to distinguish crime on a property's own block from broader neighborhood conditions. A separate event-study literature achieves much finer spatial resolution by linking individual properties to discrete, point-located events, including sex-offender arrivals (Linden and Rockoff, 2008; Pope, 2008a) and methamphetamine-lab discoveries (Congdon-Hohman, 2013), but typically focuses on a single crime or event type. Point-located designs have also been applied to murders in Sydney (Klimova and Lee, 2014), and tract- or ward-level panels to violent and property crime in the San Francisco Bay Area, Barcelona, Stockholm, Columbus, and Jacksonville (Bishop and Murphy, 2011; Buonanno et al., 2013; Ceccato and Wilhelmsson, 2011; Tita et al., 2006; Lynch and Rasmussen, 2001); at the city level, crime predicts out-migration and price declines (Cullen and Levitt, 1999; Besley and Mueller, 2012; Ajzenman et al., 2015). Across these designs the estimated effect of crime is usually negative, though not for every offense (Table OA4). Studies that separate offenses report zeros or positive estimates for some property crimes: Braakmann (2017) for burglary and vehicle crime, which he attributes to reverse causality, and Taltavull de La Paz et al. (2022) for property crime, vandalism, and non-aggravated assault committed farther from the home. I combine this spatial resolution with broad offense coverage, linking incidents across twenty-two offense categories to individual properties and estimating the price effect of an additional incident of each offense under a common design. This breadth allows me to ask not only which offenses are priced, but why. Gibbons (2004) conjectures that hard-to-observe crimes have weak price effects because prospective buyers rarely learn of them; this paper provides the first systematic test of that conjecture's implications, across twenty-two offenses under a common design. Because offenses differ simultaneously in severity, exposure, visibility, and persistence, comparisons across offenses alone cannot establish which dimension drives capitalization. I therefore examine whether the size of the discount tracks an offense's ability to predict its own recurrence and exploit within-offense variation in where incidents occur to probe the role of observability (Table 3, Appendix Table A5, and Tables OA10 and OA12).

Second, I trace how crime affects the housing transaction rather than observing only the clearing price. I examine asking prices, time on market, withdrawal, and the sale-to-list price ratio, allowing me to distinguish whether crime is incorporated into the seller's initial asking price or emerges through subsequent market adjustment (Table 4). Boggess et al. (2013) find that higher crime is followed by higher housing turnover, and Wentland et al. (2014) estimate price and time-on-

market effects of registered sex offenders from MLS data; neither traces where in the transaction the adjustment occurs or does so across offenses.

Third, I examine how crime is capitalized in rental transactions, a market studied far less in the crime-capitalization literature. Existing rent studies cover few offenses. Most use aggregate crime rates for community areas (Rizzo, 1979), cities (Koirala et al., 2023), or labor-market locations (Donovan et al., 2024), often within compensating-differentials frameworks. Studies at the property level examine one or two offense types: Carrillo and Yezer (2021) show theoretically that rent and value hedonics can diverge and, in an application, compare responses to homicide and violent crime within 0.25 miles of individual units, and Klimova and Lee (2014) link Sydney rental listings to nearby murders and find no effect on rents. Relative to these studies, I link rental listings to incidents across twenty-two offense categories and trace crime through the margins of both markets (asking prices and rents, time on market, withdrawal, and closing prices) within one city and design (Tables 4 and 6).

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework, which links crime capitalization to the information available to market participants and to belief updating about exposure relevant to the housing transaction. Section 3 describes the property-sale, rental-listing, and geocoded crime data and the construction of property-level crime exposure. Section 4 presents the empirical strategy for estimating the effect of an additional nearby crime incident on housing-market outcomes. Section 5 reports the main results, examines heterogeneity across offenses and incident locations, and assesses robustness. Section 6 interprets the results: why asking prices respond little, who carries the information, what the estimates mean for welfare, and how they bear on valuing crime reduction. Section 7 concludes.

2 Conceptual framework

The observed crime discount is the price consequence of a change in expected crime exposure. To make this explicit, suppose first that a household values occupancy at a property over a fixed horizon (thereby allowing us to ignore resale value). Let q_k denote the buyer's subjective expected number of future incidents of offense k near the property over that horizon, and let w_k denote willingness to pay to avoid one expected incident, expressed as a fraction of the property's value before the incident. Expected crime exposure therefore carries value $w_k q_k$.

Buyers enter the market with prior beliefs about q_k, formed from information already available about the location. A newly realized incident can affect valuation through two additional margins. First, information about the incident must reach the relevant market participant. Let $S_k \in [0, 1]$ denote the probability of information arrival: $S_k = 0$ means that the participant never learns of the incident, while $S_k = 1$ means that information about it reaches the participant with certainty. Second, conditional on information arrival, the incident may change the participant's subjective

expectation of future exposure. Define

$$b_k(S_k) \equiv \frac{\partial q_k}{\partial C_k^o}, \tag{1}$$

where C_k denotes realized incidents of offense k and C_k^o those whose information reaches the participant. Thus $b_k(S_k)$ measures the change in expected future incidents generated by one additional observed incident. Writing b_k as a function of S_k allows the update per observed incident to differ across offenses with how often such incidents are learned of. Housing markets price what buyers can observe: asymmetric information about a property's condition and its surroundings shows up in prices and in who buys (Garmaise and Moskowitz, 2004; Kurlat and Stroebel, 2015; Stroebel, 2016), and disclosure of a hazard moves prices only once buyers learn of it (Pope, 2008b; Bernstein et al., 2019; Hino and Burke, 2021; Bakkensen and Barrage, 2022; Gourevitch et al., 2023). In the crime setting, Gibbons (2004) argues that visible offenses such as vandalism are priced because they are hard to conceal from buyers, while information about other crimes stays with sellers and neighbors. The availability of information does not ensure that it enters a decision. Chetty et al. (2009) show that consumers respond more strongly to taxes included in posted prices, even when they know that additional taxes apply. Bordalo et al. (2012, 2013) show how salient payoffs and attributes receive disproportionate weight in choice. These mechanisms motivate distinguishing information arrival from the response to information, but salience need not operate through belief updating alone. The comparisons below do not separately identify information arrival, belief updating, and valuation.

Unlike S_k, $b_k(S_k)$ is not bounded between zero and one. If $b_k(S_k) = 0$, the participant learns about the incident but does not revise expected future exposure. A value between zero and one means that one additional realized incident raises expected future exposure by less than one incident; a value of one implies a one-for-one increase; and a value greater than one means that one realized incident leads the participant to expect more than one additional future incident over the relevant horizon. Importantly, this is a subjective belief update, not the objective recurrence rate of the offense.

The update concerns counts, not value. A buyer can expect more incidents and attach no value to avoiding them, so $b_k > 0$ with $w_k = 0$ and $\kappa_k = 0$. This is why relevance and updating enter as separate terms.

Following Rosen (1974), define the per-incident discount as the implicit price of an incident: the proportional change in the value of the property, P, per additional realized incident of offense k nearby, C_k,

$$\kappa_k \equiv -\frac{\partial \ln P}{\partial C_k}. \tag{2}$$

This is the capitalization effect in the terminology of Kuminoff and Pope (2014).[2] The framework

[2] κ_k is the marginal effect of one additional incident at the property's current exposure. The linear specification in Section 4 estimates a regression-weighted average of this effect across counts, with weights set by the residual variation in exposure; Table OA6 reports how it varies with the count.

assumes that it equals the product of three terms,

$$\kappa_k = \underbrace{w_k}_{\text{WTP to avoid one expected incident}} \times \underbrace{S_k}_{\text{information arrival}} \times \underbrace{b_k(S_k)}_{\text{expected incidents per observed incident}}, \quad (3)$$

where w_k captures the location-specific value of avoiding expected exposure to offense k. An offense may impose substantial costs on its victims yet carry little location-specific valuation if those costs are not associated with occupying or owning property at that location.

Written in units, w_k is a fractional price change per expected future incident, S_k is the share of realized incidents that reach the buyer, and $b_k(S_k)$ is expected future incidents per observed incident, so the three terms multiply to the log-price discount per realized incident estimated empirically:

$$\underbrace{\frac{\Delta \ln P}{\text{expected incident}}}_{w_k} \times \underbrace{\frac{\text{incidents observed}}{\text{incidents realized}}}_{S_k} \times \underbrace{\frac{\text{expected incidents}}{\text{observed incident}}}_{b_k(S_k)} = \frac{\Delta \ln P}{\text{realized incident}}. \quad (4)$$

Consider a buyer who would pay 1 percent of a property's value to avoid one expected future burglary, so $w_k = 0.01$. If one realized burglary reaches the buyer ($S_k = 1$) and raises expected future burglaries over the horizon from 0.5 to 2.5, the belief update is $b_k(S_k) = 2$ and, since the incident is observed ($S_k = 1$), the implied discount is $\kappa_k = w_k\, b_k(S_k) = 0.01 \times 2 = 0.02$, or 2 percent per realized burglary. The framework expresses κ_k as a positive magnitude. The estimate δ_k in the empirical sections is a signed price change, so a 2 percent discount appears as $\delta_k = -2$. A value of $b_k(S_k)$ above one simply means one realized burglary leads the buyer to expect more than one additional future burglary; it is not a probability. Had the incident not reached the buyer, or reached the buyer without changing expected exposure, this channel would generate no price response.

A zero observed discount can therefore arise through three distinct channels: willingness to pay to reduce expected exposure may be zero ($w_k = 0$); information about the realized incident may not reach the participant ($S_k = 0$); or the incident may be fully observed but leave expected future exposure unchanged ($b_k(S_k) = 0$). The last case is particularly important. A buyer may place a high value on avoiding future homicide and learn that a homicide occurred nearby, yet view the incident as isolated and therefore not revise the expected risk of future homicide. The price response is then zero despite high willingness to pay and full information. Thus, full information ensures that the incident is known, but not that it changes beliefs about future exposure; a zero capitalization estimate therefore need not imply low willingness to pay.

Equation (3) is an accounting decomposition, not a structural model. Equation (3) assumes that the buyer's entire valuation change is reflected in the sale price. With the seller's reservation value fixed, this corresponds to $\theta = 1$ in the bargaining illustration in Section 6.4, which relaxes this assumption. Prior hedonic estimates of crime report the price response to a realized incident and interpret it as a willingness to pay. With $S_k = 1$ and $b_k = \rho_k$, the discount per realized incident

is $w_k \rho_k$: willingness to pay per expected future incident, scaled by how many such incidents one realized incident implies. It equals w_k only when $\rho_k = 1$. The decomposition here keeps the two apart and asks what the data say about each term. Kuminoff and Pope (2014) note a further wedge even then, re-sorting of heterogeneous buyers and sellers, so a capitalization effect recovers marginal willingness to pay only absent that sorting (Rosen, 1974; Bishop and Murphy, 2011). The tests in Section 5.2 use only that the three terms are separable.

Prediction 1: Observability and Relevance. An incident is capitalized only if it reaches the buyer and bears on what the buyer is purchasing ($S_k > 0$ and $w_k > 0$). Holding the offense fixed, incidents that are more likely to be observed should carry larger discounts than incidents that go unseen. Across offenses, only those whose threat stays with the location should price; an offense whose harm travels with the household should not, however observable it is. I test the first condition by splitting incidents of the same offense by where they occurred and whether they are visible from outside the dwelling, and the second by classifying offenses by what the harm attaches to.

Prediction 2: Recurrence. The belief update $b_k(S_k)$ is subjective and is not observed directly. I therefore ask whether capitalization tracks how well each offense predicts future crime. Holding valuation and information arrival fixed, stronger predictive content should generate a larger discount; across offenses these other components may differ, so recurrence alone need not determine the ranking of discounts. I use a forecasting regression of whether current crime predicts recurrence to measure what an incident objectively implies about future crime, not what buyers believe it implies. If discounts do not rise with recurrence, recurrence alone does not determine which offenses are priced.

Prediction 3: Price Formation. The framework describes a change in buyers' valuation but does not say where in the transaction that change appears. If sellers anticipate the effect on buyers' willingness to pay, asking prices should adjust. If they do not, the discount can still emerge after listing, through a lower sale-to-list ratio, a longer spell on the market, or a greater chance the listing is withdrawn. I use these margins to trace where capitalization enters the transaction.

3 Data

The analysis combines three sources: residential listings from the Chicago Multiple Listing Service, incident-level crime reports from the Chicago Police Department, and measures of neighborhood conditions matched to each parcel. This section describes each source and the construction of the estimation sample.

Chicago is well suited for this analysis because it provides variation along both dimensions the design requires. First, the volume and breadth of crime incidents allow effects to be estimated

separately across the 22 offense categories reported here: the data contain 10,766 homicides alongside more than 1.3 million thefts over the sample period. Second, crime varies substantially at fine spatial scales: roughly 55 percent of the variation in doorstep crime exposure occurs within tract-months, against 14 percent for crime in the surrounding 0.20–0.30 mile ring. Doorstep crime therefore provides substantially more within-neighborhood variation than crime in the surrounding ring.

Housing transactions. The housing data are the universe of Chicago Multiple Listing Service (MLS) closed transactions in single-family, condominium, and residential-income properties. I keep transactions closing between 2007 and 2025 with a recorded close price between $10,000 and $10,000,000, and parcels whose recorded parcel count is the same at every sale, which excludes properties that were split or consolidated between sales. The resulting sample is 530,395 sales on 337,461 parcels, of which 136,866 parcels sell at least twice (329,800 sales) and so identify the within-property comparison. Single-family homes make up 88 percent of these repeat sales, two-to-four-unit buildings 11 percent, and condominiums 1 percent. Table OA1 reports the construction. Each record carries the property identifier, close date, and close price and the full listing history: original and revised list prices, days on market, and whether the listing ended in sale or withdrawal. The sale-price sample consists of closed transactions; the withdrawal analysis instead uses all MLS listings, whether or not they ended in a sale. That history lets me trace how crime affects asking prices, price revisions, time on market, and eventual transaction outcomes.

Lease listings. The rental analysis uses MLS lease listings subject to the same property-type, location, and parcel-count filters as the sale sample: single-family, condominium, and residential-income units in Chicago, on parcels with a constant parcel count, listed between 2007 and 2025 with monthly asking rents of $500 to $10,000. These restrictions yield 191,155 lease listings. Only 0.1 percent record a closing rent, so the rental price outcome is the asking rent, with crime exposure measured relative to the list date. The design otherwise mirrors the sale analysis, including days on market and withdrawal. Applying the same near–far comparison to leases allows me to examine whether the offenses priced in sales also affect asking rents.

Property and listing characteristics. Table A1 summarizes both samples. The median sale property has three bedrooms and two bathrooms and closes at $279,000; the median lease is a smaller, two-bedroom unit with a monthly asking rent of $1,950. Physical characteristics (living area, bedrooms, bathrooms, and year built) come from standardized listing fields and are available for the large majority of transactions. Two-thirds of sale listings record at least one asking-price reduction before sale.

Crime incidents. Incident-level crime data come from the Chicago Police Department's public "Crimes, 2001 to present" file. Over 2007–2025, 5,657,404 offenses were reported citywide, of which 98.8 percent carry coordinates for the location of the incident; coverage varies by offense, from

78.8 percent for embezzlement to above 99 percent for most offenses (Table 1). Figure OF1 maps their density across the city. Incident locations are released at block-face resolution: the published point is shifted from the actual location for partial redaction but falls on the same block, so distance between a parcel and an incident is measured with error of up to a block face. For homicide the file records one row per victim rather than per incident, so homicide counts, and the per-incident homicide estimates below, are per victim record.[3]

I estimate separate effects for 27 of the 29 offense groups, omitting human trafficking and nonforcible sex offenses (222 incidents combined) as too sparse.[4] Of these 27, I report 22 in the main analysis. I set aside the residual "all other offenses" category, which names no specific offense, and four offenses, kidnapping, embezzlement, extortion, and child pornography, whose estimates are too imprecise to be informative.[5] The twenty-two reported offenses appear at every window in Appendix Table A4.

For each listing I count incidents of each offense type within concentric rings of the parcel centroid over pre-transaction windows of one to twelve months, using a ball-tree spatial join,[6] and time the counts to the listing's close date (to its list date on the lease side). Table 1 reports, for each offense, its citywide count, geocoding rate, and near exposure, together with the raw cross-sectional price gap between listings with and without a nearby incident. These raw gaps are large, exceeding 40 percent for homicide and narcotics, and reflect differences between blocks with and without nearby crime rather than the effect of crime itself. Listings with an incident within 0.05 miles in the three or six months before sale are also cheaper, smaller, and slower to sell (Table OA2).

Neighborhood conditions. To guard against the discount proxying for physical decline or disinvestment rather than crime, I assemble five predetermined measures of neighborhood condition at the parcel: tree canopy, active business licenses, 311 service requests, vacant buildings, and building permits.[7] Each enters with the same near-ring and far-annulus structure as crime, differenced

[3]To protect victim privacy, the Chicago Police Department releases each incident's address "at the block level only" and does not identify the specific location (City of Chicago Data Portal, *Crimes – 2001 to Present*); the recorded point is placed on the correct block, so it can differ from the true location by up to a block face. Chicago blocks are about 330 feet east–west and 660 feet north–south, against a near-ring radius of about 264 feet, so the shift can move an incident into or out of the near ring; it adds measurement error to the near count, which is expected to attenuate δ toward zero.

[4]The pornography/obscene-material group is predominantly child pornography rather than adult material: about two-thirds of the Chicago Police incidents so classified are child pornography (IUCR 1582) or related offenses involving children, and roughly one-quarter is adult obscene material (obscene telephone calls are classified separately, under all other offenses). I therefore label this group "child pornography/obscenity" throughout. It is sparse, returns nulls in every market, and its estimates appear only in the appendix.

[5]Their standard errors run from roughly 0.6 to 1.3 percentage points, too wide to rule out either a null or a large discount.

[6]Distances are great-circle (haversine) distances between each incident's coordinates and the parcel's latitude–longitude centroid (the Cook County Assessor coordinate). A ball-tree is a spatial index that sorts the geocoded incidents by great-circle distance so that all incidents within a given radius of a parcel can be retrieved efficiently; the resulting ring counts are exact, identical to computing every parcel–incident distance directly.

[7]Tree canopy is percent canopy cover from the U.S. Geological Survey National Land Cover Database (NLCD) Tree Canopy Cover product (30 m resolution). Business licenses, 311 service requests, and building permits are geocoded records from the City of Chicago Data Portal (`https://data.cityofchicago.org`). Before the City's unified 311 system launched in December 2018, the portal releases requests in category-specific historical files; the 311 control

Table 1: Crime Incidence and Near Exposure

Offense	Incidents (city)	Geocoded (%)	Mean near count	% any near	Raw price gap (%)
Theft	1,364,912	97.8	5.56	83.2	13.7
Assault (non-domestic)	739,961	99.7	1.92	56.3	-20.9
Assault (domestic)	670,887	99.7	1.71	55.5	-35.6
Criminal damage/vandalism	632,166	99.6	1.88	67.1	-18.2
Narcotics	439,526	97.9	0.83	25.3	-45.5
Other offenses	360,851	99.0	1.09	49.5	-24.1
Burglary	296,724	99.6	0.97	47.1	-15.5
Motor vehicle theft	292,290	99.6	0.87	46.5	-9.9
Robbery	213,364	99.8	0.55	32.3	-12.2
Trespass	138,862	99.7	0.41	23.8	-1.3
Fraud	136,957	97.5	0.71	33.4	18.0
Weapons	114,818	99.6	0.23	14.4	-36.1
Sex offenses	63,549	89.2	0.18	14.6	-9.8
Disorderly conduct	39,056	99.7	0.09	7.4	-24.2
Forgery	32,711	97.1	0.11	8.9	15.9
Prostitution	31,173	99.6	0.07	2.2	-14.9
Family (nonviolent)	29,131	99.0	0.07	5.9	-35.2
Homicide	10,766	100.0	0.02	2.0	-42.9
Arson	9,488	99.7	0.02	2.1	-35.8
Bad checks	9,424	97.3	0.03	3.0	12.8
Gambling	8,105	99.4	0.02	1.2	-72.6
Liquor violations	7,941	99.5	0.03	2.2	2.5
Kidnapping	5,921	99.5	0.01	1.4	-34.0
Extortion	3,281	98.7	0.01	1.1	4.4
Child pornography/obscenity	2,677	89.8	0.01	0.6	-4.3
Stolen property	1,586	98.6	0.00	0.4	1.3
Embezzlement	1,055	78.8	0.00	0.4	45.4

Notes: Offense groups ordered by citywide incident count (Chicago Police Department, 2007–2025). Near exposure: 0–0.05 miles, twelve months before sale (N = 530,395). Raw gap: cross-sectional log-price difference (percent) between listings with and without a near incident; descriptive only.

within parcel and tract×month, but as a cumulative stock of events recorded more than twelve months before the close, that is, before every crime-exposure window used, so it is predetermined relative to the crime and cannot be an outcome of it. Permits, licenses, 311 requests, and vacant buildings are time-varying, counted as of that pre-crime cutoff for each listing; tree canopy is read from the land-cover raster two years before the sale. Foreclosures lower nearby prices and raise nearby crime (Campbell et al., 2011; Ellen et al., 2013; Anenberg and Kung, 2014), so I control for the predetermined vacancy stock near and far.

4 Empirical Strategy

4.1 Design and Identification

The design is a spatial difference-in-differences with two-way fixed effects and continuous treatment. Crime exposure within a narrow ring around the property constitutes the near treatment, while exposure in a wider annulus around the same property provides the spatial comparison. Identification combines three comparisons: within parcel across transactions, within tract and month, and near relative to farther crime exposure (Figure 1). Parcel fixed effects absorb permanent differences across properties, and tract-by-month fixed effects absorb contemporaneous shocks shared by the surrounding neighborhood. This division of variation is borne out in the data: only 14 percent of far-ring crime exposure varies within tract-months, so the surrounding annulus is almost entirely a neighborhood-level quantity that the fixed effects absorb nearly in full, whereas a majority of doorstep exposure survives within tract-months (Section 3). The near-relative-to-far contrast is therefore identified from within-neighborhood differences in exposure. The parameter of interest is the differential price response,

$$\delta = \beta_{\text{near}} - \beta_{\text{far}}.$$

The estimate δ_k corresponds to $-\kappa_k$ in Equation (3) under the linear approximation and the far-ring restriction: it is a regression-weighted average of the per-incident effect across counts.

combines those files for 2011–2018 (graffiti removal, sanitation code complaints, vacant and abandoned buildings, and abandoned vehicles) with requests of all types from the unified system for 2019–2024, so its composition changes in 2019. Vacant and abandoned buildings are "Vacant and Abandoned Buildings Reported" records from the same portal; because such complaints also enter the City's unified 311 stream from 2019 onward, this measure partly overlaps the total-311 control over the recent period.

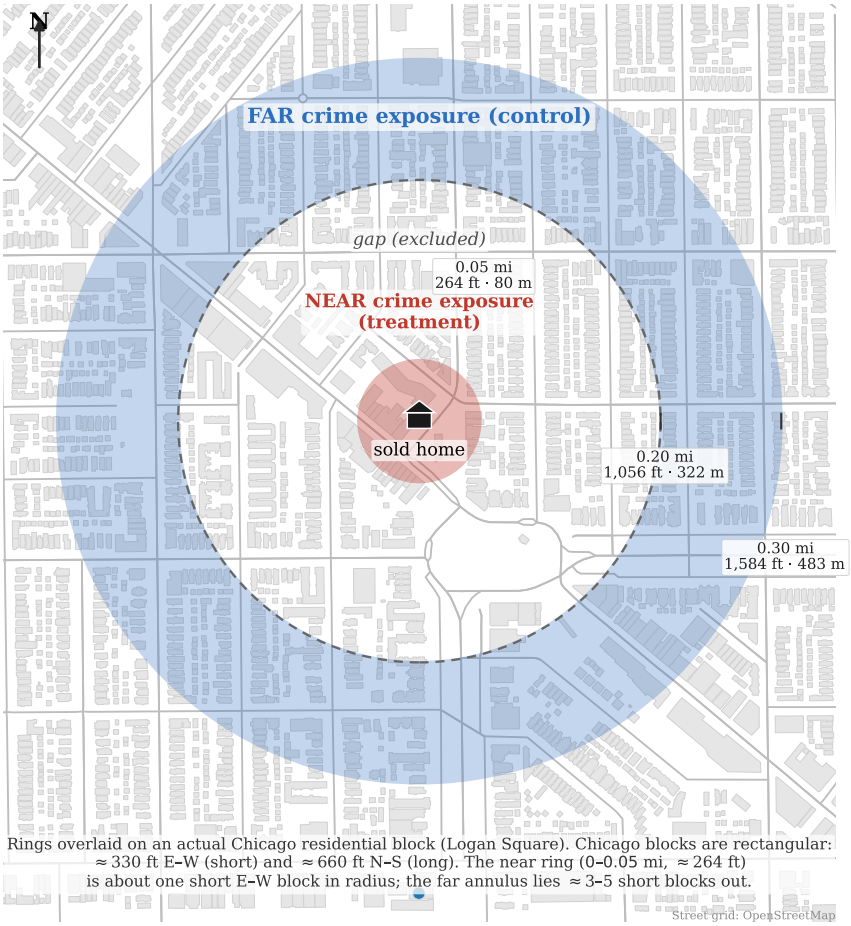


Figure 1: The near–far design. Crime within the near disk (0–0.05 miles, about 260 feet, roughly one short east–west Chicago block) is the treatment; crime in the far annulus (0.20–0.30 miles, about three to five short blocks out) is the spatial comparison, separated from the doorstep by a gap so the two rings are non-adjacent. The near–far difference $\delta = \beta_{\text{near}} - \beta_{\text{far}}$ differences out conditions common to the surrounding neighborhood. Near disk in red, far annulus in blue; incidents in the gap and beyond the far ring (grey) are excluded.

For property i transacting at date t, let Near_{it} be the count of incidents of a given offense within 0.05 miles of the parcel in the m months before the sale, and Far_{it} the count in the 0.20–0.30 mile annulus over the same window, the same comparison ring as Tang and Le (2023); Appendix Table A2 reports alternative far rings. Stale counts SNear_{it} and SFar_{it}, near- and far-ring incidents in the part of a twelve-month lookback that falls *beyond* the m-month window, and therefore non-overlapping with Near_{it} and Far_{it}, absorb slower-moving local conditions; they are empty when $m = 12$. The window length m is not fixed a priori; I estimate across $m = 1, \ldots, 12$ months and examine the choice in Section 4.2. The estimating equation is

$$\ln P_{it} = \beta_{\text{near}}\text{Near}_{it} + \beta_{\text{far}}\text{Far}_{it} + \gamma_n \text{SNear}_{it} + \gamma_f \text{SFar}_{it} + \mu_i + \lambda_{c(i)t} + \varepsilon_{it}, \tag{5}$$

where μ_i is a parcel fixed effect and $\lambda_{c(i)t}$ a tract×month fixed effect.

Identification requires that, after accounting for parcel and tract-by-month fixed effects and

the included controls, the difference in crime exposure between the near ring and the surrounding annulus is unrelated to other factors that affect the sale price. This does not require near and far rings to be alike in levels, or unconditional parallel trends across high- and low-crime properties. Parcel fixed effects absorb time-invariant property differences, and tract-by-month fixed effects absorb shocks shared within a tract and month. What remains is block-level variation across a property's own transactions. Formally, let $\mathbf{X}_{it} = (\text{Near}_{it}, \text{Far}_{it})'$. Identification requires

$$E\left[\widetilde{\mathbf{X}}_{it}\,\widetilde{\varepsilon}_{it}\right] = \mathbf{0}, \tag{6}$$

where tildes denote residualization with respect to the fixed effects and controls in Equation (5). Under this condition β_{near} and β_{far} are identified, and so is $\delta = \beta_{\text{near}} - \beta_{\text{far}}$.

Two threats to this assumption, and the interpretation of δ, are addressed directly.

1. *No shocks correlated with the near–far difference.* The concern is an event that raises crime on the block relative to the surrounding annulus and lowers the price for a reason of its own, such as a building abandoned next door. A shock shared by the block and the annulus is differenced out; one confined to the block is not. Such an event is a confounder: it moves both the near–far crime difference and the price, so the estimate would attribute its price effect to crime. Five predetermined neighborhood measures (building permits, vacant buildings, business licenses, tree canopy, and 311 service requests), each entered as its own near–far difference and dated before the crime window, proxy for the block-specific events that can be observed (Section 5.6, Appendix Table A7).

2. *No reverse causality or mean reversion.* Crime exposure is measured before the sale, so the price cannot mechanically produce it. But a property that sells at a temporarily depressed price may sit on a block where crime was already rising, so the estimate would pick up the block's trajectory rather than the incident. Crime that occurs after the closing cannot affect a price already set, so a post-sale placebo that loads would indicate this kind of confounding; a null is consistent with the assumption without establishing it. Every estimate has a mirror placebo that substitutes post-sale for pre-sale crime.

The interpretation requires that the far ring is beyond the effect's reach. δ compares marginal responses to near and far exposure and equals the doorstep effect only under the far-ring restriction, that crime a fifth to three tenths of a mile out has no doorstep effect of its own. Crime at that distance is the benchmark for what would have happened on the block anyway; if an incident there still moved the price, the contrast would recover a relative effect rather than the full doorstep effect. The spatial decay results are consistent with the restriction: the discount is gone by the adjacent block.

The design does not assume that crime is randomly distributed across the city, across neighborhoods, or over time; only that the residual near–far variation is unrelated to unobserved price shocks. Section 5.6 reports each check.

The parcel fixed effect restricts identification to the 136,866 parcels that transact more than once (329,800 sales). The tract-by-month fixed effect then drops any transaction that is the only sale in its cell, together with the singletons this creates, leaving the 242,815 transactions that identify the two-way model and form the sample in the results tables (Table OA1).

Enforcement intensity affects recorded crime directly (Levitt, 1998) and can bias the estimate in either direction. Where enforcement deters crime and improves the block, detection may still raise recorded counts, so a block that is getting safer shows more incidents and the discount is attenuated. Where enforcement is aimed at blocks that are already deteriorating, recorded counts rise alongside falling prices and the discount is inflated. Tract-by-month fixed effects absorb enforcement changes shared within each tract and month.[8] Uneven deployment within tracts, including enforcement targeted at particular addresses, remains a concern. This matters for offenses recorded through enforcement rather than victim complaint, which here means narcotics, prostitution, gambling, and liquor violations, and I describe those measures as enforcement-recorded offenses rather than as a census of crime.

4.2 Choice of Spatial Bands and Exposure Windows

The rings and windows in (5) are chosen from the data rather than imposed. I estimate the price effect as a smooth function of distance and time since incident by a tensor-product spline[9] (Wood, 2017; Korolev, 2026). The distance profile is significantly negative at the doorstep and returns to zero by the adjacent-block annulus, so the near ring sits in the steep part of the profile and the far ring beyond its reach (Figure 2; Online Appendix J). Because administrative units are not the area over which a household experiences crime, aggregating to them would make coefficients shift with the unit chosen, the modifiable areal unit problem (MAUP), so I report estimates across nine near–far ring definitions (Appendix Table A2) and all one-to-twelve-month windows. Tables that report a single number per offense use the central window of the range over which that offense's discount is significant at five percent (pooled sample, headline 0–0.05 versus 0.20–0.30 mile ring), taking the lower of the two central windows when that range has even length, and offenses with no significant window are shown at the median window, six months. Summary tables use the central window among those with significant discounts; Table 2 reports estimates across all twelve windows. Significant-window counts are descriptive; the windows are nested, not independent.

Section 5.6 reports four robustness checks. A post-transaction placebo addresses reverse causality and mean reversion; predetermined near–far measures of neighborhood conditions address hyper-

[8] A Chicago police beat averages about 0.8 square miles and spans roughly three census tracts; the 791 tracts in the estimation area average about 0.29 square miles. Beats nest within sectors and sectors within the city's roughly two dozen districts. Beat and tract boundaries cross rather than nest, so beat-level deployment is absorbed only in part.

[9] A tensor-product spline models a function of two variables as a weighted sum of products of one-dimensional basis functions, here a basis in distance from the parcel and a basis in months since the incident. Each product term lets the profile in one dimension vary with the other, so the distance profile can differ across lags without imposing a parametric form. The basis sizes are $c(6, 12)$, six in distance and twelve in lag, and the smoothness penalty is chosen by restricted maximum likelihood (Wood, 2017).

local confounding; a full-universe model of sale propensity and a complementary selection correction assess endogenous transaction selection; and a mass-point diagnostic following Caetano (2015) and Caetano et al. (2021) addresses selection into zero crime exposure. The full set of diagnostics is reported in the robustness section. The comparisons that follow use incident location and recurrence to distinguish among the mechanisms in the conceptual framework.

The parameter of interest is the effect of a recent incident on the property's block, relative to the surrounding neighborhood in the same month. The spatial scale is not imposed: the distance profile in Figure 2, estimated without rings, locates the response at the doorstep, and the rings follow it. The estimates do not capture the long-run co-movement of crime and prices across neighborhoods.

5 Results

5.1 Sale Prices

Appendix Table A4 reports estimates for every offense at all twelve windows.

Table 2 reports sale-price estimates for all twenty-two offenses. Across the twelve windows, the estimates fall into four broad groups: *priced*, *suggestive*, *precise nulls*, and *inconclusive*. Table 2 is ordered accordingly, with classification criteria reported in the table notes.

Homicide, burglary, narcotics, and vandalism are priced. Disorderly conduct and sex offenses are suggestive. Twelve offenses are precise nulls: their intervals rule out a one-percent discount in at least nine of twelve windows, and for larceny, motor vehicle theft, and non-domestic assault they also rule out a discount as large as burglary's (0.24 percent per incident); the estimates for bad checks and family offenses are positive. Weapons, arson, gambling, and stolen property are inconclusive.

Table 2: Sale Prices

Offense	Identifying parcels	Sig. discount windows	Min δ (s.e.) [mo]	Max δ (s.e.) [mo]	Peak δ (s.e.) [mo]
Priced					
Homicide	5,701	2/12	-5.86^{*} (3.35) [1]	-1.24 (1.43) [8]	-3.15^{**} (1.38) [11]
Narcotics	53,844	5/12	-0.93 (0.75) [1]	-0.18 (0.19) [6]	-0.38^{**} (0.18) [8]
Burglary	88,094	6/12	-0.34^{**} (0.14) [6]	-0.18^{*} (0.09) [12]	-0.34^{**} (0.14) [6]
Vandalism	104,661	2/12	-0.17^{**} (0.08) [6]	0.26 (0.22) [1]	-0.17^{**} (0.08) [6]
Suggestive					
Disorderly conduct	19,550	1/12	-1.33^{*} (0.78) [2]	-0.46 (0.54) [4]	-0.65^{**} (0.29) [12]
Sex offenses	36,393	1/12	-0.47^{**} (0.23) [12]	0.11 (0.35) [3]	-0.47^{**} (0.23) [12]
Precise nulls					
Larceny/theft	115,548	0/12	0.01 (0.02) [12]	0.08^{***} (0.03) [1]	–
Motor vehicle theft	85,781	0/12	-0.02 (0.09) [12]	0.51^{*} (0.30) [1]	–
Assault (non-dom.)	91,783	0/12	-0.05 (0.06) [7]	0.01 (0.11) [3]	–
Bad checks	8,919	0/12	0.66^{*} (0.35) [11]	2.20^{**} (1.03) [1]	–
Family (non-violent)	15,928	0/12	1.00^{*} (0.51) [12]	2.16^{***} (0.82) [5]	–
Fraud	67,030	0/12	-0.36 (0.23) [1]	-0.04 (0.09) [6]	–
Assault (domestic)	90,500	0/12	-0.19^{*} (0.10) [12]	0.02 (0.15) [5]	–
Trespass	54,025	0/12	-0.15 (0.14) [6]	0.18 (0.22) [3]	–
Robbery	65,876	0/12	-0.06 (0.14) [9]	0.30 (0.42) [1]	–
Forgery	23,614	0/12	-0.07 (0.59) [1]	0.47 (0.45) [2]	–
Liquor law	6,393	0/12	0.08 (1.20) [1]	1.34 (0.91) [2]	–
Prostitution	6,326	0/12	-1.00 (0.82) [1]	0.08 (0.12) [12]	–
Inconclusive					
Weapons	32,294	0/12	-0.62 (0.47) [5]	0.11 (0.58) [3]	–
Arson	6,116	0/12	-0.20 (1.06) [11]	1.27 (3.58) [1]	–
Gambling	3,125	0/12	-4.77 (4.33) [3]	1.99 (8.32) [1]	–
Stolen property	1,262	0/12	-2.23 (2.63) [2]	0.81 (1.20) [8]	–

Notes. $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$: percent change in sale price per incident within 0–0.05 miles, relative to 0.20–0.30 miles. Parcel and tract × year-month fixed effects; standard errors clustered by tract; $N = 242{,}815$ repeat sales, 2007–2025. Estimated at each cumulative window, one to twelve months. Sig. windows: negative and significant at 5%. Min δ and Max δ: the most negative and most positive estimates across the twelve windows (standard error; window in brackets). Peak: window of the most negative significant estimate. Identifying parcels: exposure varies within parcel, counted at the twelve-month window. Priced: a significant discount at two or more windows, with adjacent windows agreeing in sign. Suggestive: a significant discount at only one window, marginally significant at adjacent windows. Precise nulls: no window shows a significant discount, and the 95 percent intervals rule out a one-percent discount in at least nine of twelve windows; for larceny, motor vehicle theft, non-domestic assault, bad checks, and family offenses they also rule out a discount as large as burglary's (0.24 percent), for the last two because their estimates are positive. Inconclusive: no window shows a significant discount, but the intervals cannot rule out a one-percent discount in most windows. Kidnapping, embezzlement, extortion, child pornography, and other offenses have too few incidents to estimate.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fraud, forgery, and bad checks impose direct monetary losses, yet none carries a detectable discount in the pooled estimates (Table OA3); embezzlement has too few incidents to estimate. Bad checks instead carry a positive and significant estimate (1.09, s.e. 0.53), positive at five percent in ten of twelve windows (Appendix Table A4). Splitting each offense's incidents by recorded location (residential, street, and commercial) leaves the picture unchanged (Table OA13). Bad checks have no detectable effect in any location category, and forgery's only significant estimate is positive and occurs at commercial addresses. Fraud is associated with a residential discount of 0.40 percent

(s.e. 0.15) that survives the post-sale placebo. This estimate is one of sixty-six location-specific comparisons, does not survive a Bonferroni correction, and appears only at the longest exposure windows, so I treat it as suggestive.

Replacing the near count in the main specification with indicators for one, two, three, and four or more incidents shows where the discount comes from. A single incident does not significantly lower the price at any window for any priced offense. Four or more burglaries lower it by 1.85 percent (s.e. 0.76) at the central window, significant in 8 of 12 windows, and four or more vandalism incidents by 0.92 percent (s.e. 0.45), significant in 3 of 12. For narcotics the four-or-more estimate is significant at one window and for homicide at none, where the cell holds only 14 listings, against 8,040 for narcotics (Table OA6).

Spatial decay. For burglary and narcotics, the estimated discount falls sharply beyond the near ring. Setting the near ring to each successive annulus while holding the far ring at 0.20–0.30 miles, the doorstep discount largely collapses by the 0.05–0.10 annulus and stays near zero through 0.15–0.20, the ring just inside the far control, with the rate of decay varying by offense across the intervening rings (Appendix Table A2). Two offenses show discounts beyond the doorstep. Burglary's discount remains significant when the near disk is widened to 0–0.10 miles (-0.14, s.e. 0.07; significant in 5 of 12 windows, with no significant post-sale placebo), but the 0.05–0.10 annulus alone is not significant, so the wider-disk estimate reflects the doorstep effect averaged over a larger area. Homicide is significant in the 0.10–0.15 mile annulus in 7 of 12 windows (months three to nine; -1.16, s.e. 0.64, at its ten-month central window), with no significant post-sale placebo, but not in the 0.05–0.10 annulus, so its effect extends a few blocks without a smooth decline in distance (Figure OF9). Crime at 0.20–0.30 carries no doorstep effect, so the far ring is untreated. Because $\delta = \beta_{\text{near}} - \beta_{\text{far}}$, the far-ring effect enters the design as a normalization, and any residual far-ring discount would make the estimate understate rather than overstate the doorstep effect.

Spline estimates. A tensor-product spline in distance and lag, estimated as a fixed-effects signal regression following Korolev (2026), traces the same decay (Figure 2): for the dense property offenses, burglary and narcotics, the effect is significantly negative at the doorstep and gone by the adjacent-block annulus; vandalism is a precise near-zero, consistent with its small per-incident discount (Table 2); and homicide's profile is imprecise: its band is an order of magnitude wider and includes zero at the doorstep, since homicide is too sparse to support a stable two-dimensional surface (Online Appendix J). In the lag direction, the doorstep effect is significantly negative over months 2.5–5.5 for burglary and 2.5–4.5 for vandalism and declines only slowly for narcotics.

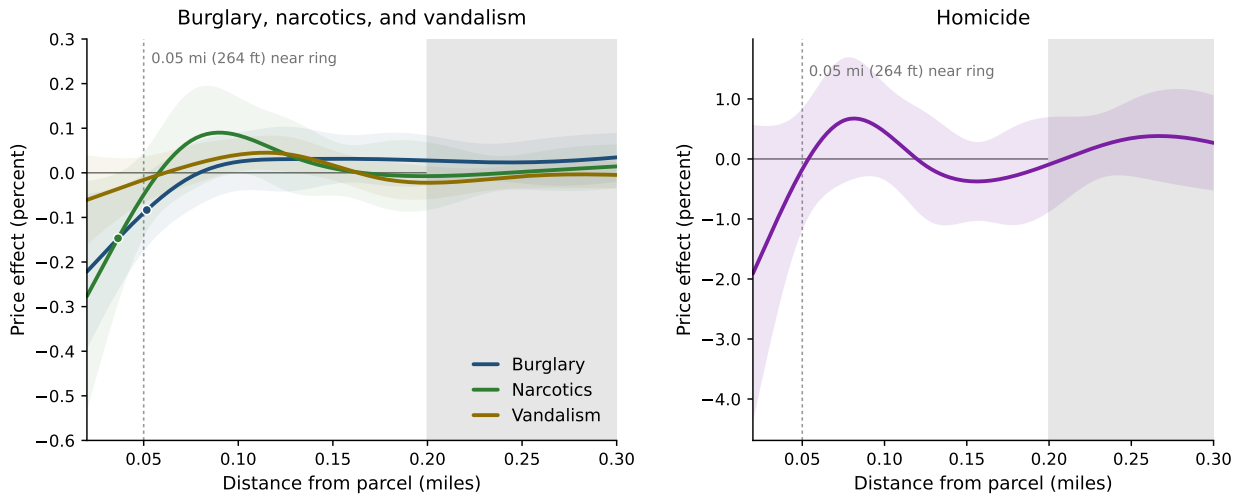


Figure 2: Price response by distance from the parcel. Distance profiles from a tensor-product spline in distance and lag, estimated as a fixed-effects signal regression in `mgcv` following Korolev (2026), fit on 0–0.30 miles and centered on their mean, averaged over the one-to-twelve-month windows; MLS close-date sample, 2007–2025, with 95 percent cluster-robust bands. *Left*: burglary, narcotics, and vandalism. *Right*: homicide, whose band is wider because incidents are rare and includes zero at the doorstep. The dotted line marks the 0.05-mile near-ring edge; the shaded band is the 0.20–0.30 mile comparison ring. The profile locates where the response is negative and where it is flat; its level is not the near–far discount in Table 2. Estimation details in Online Appendix J.

Time horizon of the effects. Estimated jointly, exposure two to six months before the sale and exposure eight to eleven months before carry different weights across offenses. Vandalism loads on the recent block and not on the late one, and the difference is significant ($p = 0.008$); narcotics loads on the late block, with the difference significant at ten percent ($p = 0.09$); for homicide and burglary the point estimates load late and early respectively, but the two blocks are not statistically distinguishable ($p = 0.12$ and 0.20; Figure OF2 and Table OA7).

The design removes most of the raw price gap near crime, and the remaining structure matters in a way that differs by offense. Table OA5 estimates the near–far contrast on the repeat-sales sample with progressively more structure. Tract-by-month effects alone reduce the raw per-incident homicide gap from about thirteen percent to about two, so most of the naive association is neighborhood and period. For the property offenses, adding observed characteristics and then parcel fixed effects shrinks the discount further, from about 0.7 to 0.2 percent for burglary and from 1.2 to 0.2 for vandalism, so a cross-sectional hedonic on these data overstates the property-crime discounts. Homicide moves the other way, from 2.3 to 3.0 percent, so the same hedonic understates it. Parcel fixed effects without tract-by-month effects give the wrong sign for homicide and double the narcotics discount, which is why the design needs both: the parcel effect removes what is permanent about the property and the tract-by-month effect removes what is moving in the neighborhood.

Wider near rings produce additional discounts that also appear in post-sale placebos. At 0–0.10 miles several offenses that show no detected discount at the doorstep produce apparent sale-price discounts for which post-sale crime also predicts the price: domestic assault, for instance, discounts by about a quarter percent in the 2009–2013 foreclosure-crisis era, while crime occurring

after the sale predicts the same discount in seven of twelve windows. That is neighborhood-level contamination, not capitalization, and the tighter doorstep ring excludes it.

5.2 Mechanisms

Sixteen of twenty-two offenses carry no detectable discount, and two more (disorderly conduct and sex offenses) are significant only at a single long-horizon window.

Location of the incident Prediction 1 says that, holding the offense fixed, incidents more likely to be observed carry larger discounts. I split incidents of the same offense in two ways: (1) by the type of place where they were recorded (a residential address, the street, or a commercial premises), and (2) among incidents at residential addresses, by whether they occurred inside the dwelling or at its frontage (the garage, porch, yard, driveway, or entryway).

The homicide and narcotics discounts come mainly from incidents on the street, and the burglary and vandalism discounts from incidents at residential addresses (Tables OA12 and OA14). Vandalism at residential addresses is associated with a discount (-0.56, significant at one percent), while vandalism in public space is not (-0.01), and the two differ significantly in seven consecutive windows, although residential and street vandalism forecast future doorstep crime at similar rates (Table OA16). Homicide's discount is concentrated in incidents on the street, although its residential and street estimates are not statistically distinguishable.

The interior–frontage split holds the offense and the address fixed and varies only what is visible from the street. Frontage also includes the porch/hallway and common-area codes, which make up 39 percent of frontage-coded incidents across all offenses (Table OA15). For burglary (Table 3), frontage incidents carry a discount (-0.74, s.e. 0.33), significant in 7 of 12 windows, while interior incidents have no detectable effect (-0.14, s.e. 0.16) at any window. The gap widens with the exposure window and is significant at twelve months ($p = 0.030$; $p = 0.109$ at the central window). The frontage estimate rests on about 45,000 incidents, and its post-sale placebo is null (-0.20, $p = 0.45$). Vandalism shows the reverse: interior incidents carry a discount (-0.73, s.e. 0.25), significant in 9 of 12 windows, compared with frontage incidents (-0.14, s.e. 0.33; Table OA18).[10] As a check that the split does not mechanically produce a difference, larceny shows no discount at either location, and the two estimates do not differ ($p = 0.686$; Table OA18).

[10]CPD masks each incident to the hundred-block, so an interior record cannot be separated between the sold parcel and a neighbor's. For burglary this matters little, since the loss is the stolen contents and any entry damage is minor; for vandalism an interior record is damage to the property itself, which lowers its value as repair cost rather than through information. I report the vandalism cells but exclude vandalism from this within-offense observability comparison.

Table 3: Two Within-Offense Tests: Observability and Relevance

	Exposure window (months before sale)											
	1	2	3	4	5	6	7	8	9	10	11	12
Panel A. Observability: burglary, residential incidents												
Frontage (visible)	−0.37	−0.45	−0.54	−0.68	−0.54	−0.74**	−0.76**	−0.74**	−0.62**	−0.70**	−0.71***	−0.73***
	(0.77)	(0.59)	(0.49)	(0.46)	(0.41)	(0.36)	(0.35)	(0.33)	(0.29)	(0.27)	(0.25)	(0.24)
Interior (hidden)	−0.17	−0.27	−0.12	−0.11	−0.20	−0.30	−0.22	−0.14	−0.18	−0.19	−0.20	−0.13
	(0.47)	(0.30)	(0.26)	(0.22)	(0.20)	(0.19)	(0.18)	(0.16)	(0.15)	(0.13)	(0.13)	(0.13)
Frontage = interior (p)	0.824	0.796	0.457	0.258	0.462	0.299	0.183	0.109	0.182	0.098	0.081	0.030
Panel B. Relevance: vandalism, frontage incidents, by target												
Damage to property	0.85	1.06	0.80	0.22	0.03	−0.49	−0.89**	−0.95**	−0.91**	−1.03***	−0.92**	−0.89***
	(1.12)	(0.83)	(0.63)	(0.58)	(0.51)	(0.47)	(0.45)	(0.43)	(0.41)	(0.39)	(0.36)	(0.34)
Damage to vehicle	0.70	0.89	0.50	0.33	0.20	0.21	0.05	−0.07	−0.13	−0.08	−0.09	0.02
	(0.93)	(0.67)	(0.56)	(0.53)	(0.48)	(0.45)	(0.42)	(0.40)	(0.40)	(0.38)	(0.37)	(0.34)
Property = vehicle (p)	0.916	0.874	0.724	0.896	0.818	0.298	0.143	0.142	0.168	0.078	0.102	0.057

Notes. δ as in Table 2, at each window; standard errors below. Panel A splits residential burglary into frontage (garage, porch, yard, driveway, entryway, porch/hallway and common areas; Table OA15) and interior incidents. Panel B splits frontage vandalism by target. Last row of each panel: Wald p for equal effects. Post-sale placebos for the frontage and interior components are significant in at most one of twelve windows; the pooled residential burglary coefficient, by contrast, is matched window-for-window by post-sale crime (Online Appendix C); both fixed-effect designs in Online Appendix D.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Target of the incident Among the priced offenses, only vandalism has sub-codes that record what was targeted, damage to the property and damage to a vehicle; homicide targets a person, and the burglary and narcotics sub-codes record the method of entry or the drug, not the target. Among frontage vandalism, damage to the house is significant in 6 of 12 windows (-0.49, s.e. 0.47, at the central window) while damage to a car is significant in none (0.21, s.e. 0.45), and the two differ at ten percent (at ten and twelve months, $p = 0.078$ and 0.057); both post-sale placebos are null (Online Appendix D).[11]

Risk of future crime Prediction 2 asks whether discounts track what an incident reveals about future crime at the property. In the framework this is the updating term $b_k(S_k)$ in $\kappa_k = w_k \times S_k \times b_k(S_k)$, the change in expected future incidents at the property once the buyer learns of an incident (Section 2). The tests below measure the objective risk that the crime data forecast, not buyers' subjective beliefs, which are unobserved; a mismatch therefore shows that prices do not track forecastable risk, not that buyers fail to update. I measure this in two ways and compare each with the discounts: (1) how well crime a few blocks away forecasts future crime at the doorstep, and (2) how strongly doorstep crime predicts its own recurrence at the same parcel.

First, crime a few blocks away forecasts crime at the doorstep: in a forecasting regression without fixed effects, past crime in the far ring (0.20–0.30 miles) significantly forecasts future crime at the doorstep (0–0.05 miles), at five percent, for twenty-one of the twenty-two offenses at the six-month horizon (all but stolen property) and for all twenty-two at twelve months, priced or not. The forecasting power is modest: once the doorstep's own past crime is controlled for, the far ring adds little (partial R^2 at most 0.09 for any offense at either horizon). The far ring thus predicts doorstep crime but is not itself priced: for the four priced offenses, doorstep crime carries a discount while far-ring crime carries none (Section 5.1). Yet how well an offense is forecast does not line up with its discount. Fraud, weapons, domestic assault, and motor vehicle theft are forecast at least as well as burglary and vandalism but carry no discount, while homicide, which carries the largest discount, is among the least forecastable (Table A5).

Second, I compute recurrence ρ by regressing future doorstep crime (after the sale) on past doorstep crime (before the sale) over windows of the same length, with parcel and tract $\times$ month fixed effects, over the same window as δ (Table OA10). If discounts tracked recurrence, offenses with higher ρ would carry larger discounts. They do not. Narcotics is the most self-predicting of the offenses in Table OA10 ($\rho = 0.40$) yet carries a discount far below homicide's; homicide's own-recurrence coefficient is small and negative ($\rho = -0.026$, s.e. 0.012) and it carries the largest. Homicide's weak recurrence is consistent with Carrillo and Yezer (2021), who find homicide far less persistent from year to year than other violent crime. That recurrence is real but small and offense-specific is corroborated by Caetano and Maheshri (2018), who find that several Dallas offenses (robbery, burglary, auto theft, and light crime) raise their own future incidence by roughly five to fifteen percent, fading within about a month, with no escalation across offense types.

[11] Because the police records mask each incident to the hundred-block, a frontage damage-to-property record can be the sold parcel itself, so this contrast carries an own-property component.

5.3 Price Formation

Prediction 3 asks where in the transaction the discount appears. I trace the effect of crime through the housing transaction. Boggess et al. (2013) find that higher neighborhood crime is followed by higher rates of housing transactions, with sales inhibited only where crime and vacancy are already high, but their data cannot say which margin adjusts. I examine asking prices, mid-listing price cuts, the sale-to-list ratio, time on market, and withdrawal, which lets me distinguish effects incorporated when the seller sets the asking price from those that emerge later in the transaction.

Table 4: Transaction Outcomes: From Asking Price to Sale

Offense	Ask Price %	Sale-to-list %	Price cut pp	Days on market %	Withdrawal pp	Close %
Homicide	0.38	0.53	$8.24^{**\ddagger}$	14.54^{**}	2.24^{**}	-3.02^{**}
	(0.57)	(0.45)	(3.30)	(6.85)	(0.96)	(1.38)
Narcotics	0.21	-0.16^{**}	0.43	−0.78	0.04	-0.29^{**}
	(0.17)	(0.07)	(0.35)	(0.98)	(0.04)	(0.14)
Burglary	−0.15	−0.01	−0.33	0.23	−0.10	-0.24^{**}
	(0.12)	(0.05)	(0.33)	(1.25)	(0.10)	(0.12)
Vandalism	0.05	0.00	−0.12	−0.10	0.00	-0.17^{**}
	(0.07)	(0.03)	(0.21)	(0.69)	(0.05)	(0.08)
Disorderly conduct	-0.43^{**}	-0.28^{**}	2.55^{***}	−1.97	0.10	-0.65^{**}
	(0.21)	(0.12)	(0.95)	(3.86)	(0.23)	(0.29)
Parcel FE	Yes	Yes	Yes	Yes	Yes	Yes
Tract × month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Near–far estimates as in Table 2, pooled 2007–2025. Columns: log asking price, log sale-to-list ratio, mid-listing price cut (percentage points), log days on market, withdrawal (percentage points), and log closing price. Closing price at each offense's central window (homicide 10, burglary 8, narcotics 10, vandalism 6, disorderly conduct 12 months); listing margins at the median significant window. Days on market uses parcel × era fixed effects; withdrawal uses all listings.

‡The price cut compares the last asking price with the original asking price of the same listing. It does not record when the cut was made, so it cannot show whether the cut followed the crime. Using the dated price-change history of each listing (available from 2015), crime during the listing does not raise the probability of a cut (Table OA23).

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Transaction outcomes For the priced offenses, the discount appears at the close rather than in the asking price (Table 4). None of the four moves the asking price, although for burglary the asking-price estimate is imprecise (−0.15, s.e. 0.12) and its interval includes the closing discount. Disorderly conduct is the only offense that lowers the asking price. Narcotics also closes below list (sale-to-list −0.16, s.e. 0.07). A nearby homicide makes the property harder to sell: time on market rises by about 15 percent, roughly six days against a median of 42, with parcel-by-era fixed effects that absorb era shifts in typical time on market, and the probability of withdrawal rises by 2.2 percentage points. Crime during a live listing does not trigger a price cut: dating each cut from the listing's price-change history (available from 2015), none of the six offenses raises the probability of a cut in the following month (homicide 0.24 percentage points, s.e. 0.35), and, as a placebo, crime in the month after, which cannot have caused the cut, has no effect either (Table OA23). Online

Appendix E reports the estimates by era.

Agent experience Prior work shows that agents' incentives and information shape transaction outcomes (Rutherford et al., 2005; Levitt and Syverson, 2008; Hendel et al., 2009; Han and Hong, 2011; Barwick and Pathak, 2015; Gilbukh and Goldsmith-Pinkham, 2024). I ask whether the asking price responds differently to nearby crime depending on the listing agent's experience in the tract, and whether the closing price responds differently depending on the buyer's agent's experience (Table 5). I proxy experience by the number of listings the agent has previously handled in the same census tract, in logs. After a nearby homicide, asking prices fall more when the listing agent is more experienced, significantly so in seven of twelve windows. Closing prices also fall more when the buyer's agent is more experienced: comparing the same agent across transactions, a one-standard-deviation increase in the buyer's agent's local experience deepens the homicide discount by 1.93 percent (significant in six of twelve windows; Appendix Table A6). For narcotics, the response appears in the asking price: it falls more with listing-agent experience in seven of twelve windows. For these offenses the discount at mean experience is not statistically significant on the smaller agent-matched sample; the response is concentrated among experienced agents.

Three further checks refine these patterns. First, the asking-price response to listing-agent experience is negative for homicide, narcotics, and vandalism in nearly every window, both with parcel fixed effects and with agent fixed effects added (Appendix Table A6). Second, for narcotics it weakens once agents listing in a tract for the first time are allowed to differ, and for vandalism it comes mainly from those first listings (Table OA24). Third, controlling for the agent's total experience (all prior listings, in any tract), the homicide results remain for both the listing agent (-0.79, s.e. 0.35) and the buyer's agent (-1.99, s.e. 0.86), and total experience itself has no significant effect on either side; for narcotics, the reverse holds: the asking price falls with the listing agent's total experience (-0.21, s.e. 0.05, significant in all twelve windows), and local experience has no effect once total experience is controlled for (Table OA25). Because agents are not randomly assigned to listings, these estimates are associations rather than causal effects.

Table 5: Agent Local Experience: Ask (Listing) and Close (Buyer)

Offense	Most negative (s.e.)	Most positive (s.e.)	Sig. windows
Panel A. Listing agent, asking price (ln list)			
Homicide	-0.97^{***} (0.34)	-0.03 (0.52)	7/12
Burglary	-0.02 (0.03)	0.06 (0.05)	0/12
Narcotics	-0.11^{**} (0.05)	-0.05 (0.08)	7/12
Vandalism	-0.06^{**} (0.02)	-0.02 (0.07)	3/12
Panel B. Buyer agent, closing price (ln close)			
Homicide	-6.27^{**} (2.60)	-0.95 (0.88)	6/12
Burglary	0.02 (0.15)	0.53^{**} (0.27)	0/12
Narcotics	-0.06 (0.47)	0.38^{*} (0.22)	0/12
Vandalism	-0.04 (0.14)	0.14^{**} (0.06)	0/12

Notes. Near–far effect interacted with agent experience, $E = \ln(1 +$ prior same-agent same-tract listings). Panel A: listing agent, asking price. Panel B: buyer's agent, closing price. Parcel, tract × month, and agent fixed effects; clustered by tract. Gradient: change in the near–far effect per log unit of E (s.d. 1.20) (negative: experienced agents price crime more), shown as the most negative and most positive estimates across the twelve windows and the number of significantly negative windows. Appendix Table A6 reports the same specification per standard deviation of E at each offense's central window. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The quantity margin Price is not the only margin on which the market can absorb a local shock. I therefore estimate the effect of each offense on the probability that a parcel is listed and on the probability that it sells (Table OA26). The listing–sale difference is the listing response minus the sale response; a positive difference means crime raises listings more than completed sales. It is not a direct estimate of withdrawal. Ten of the twenty-two offenses show a positive difference, including every offense that is priced, and the offenses that are null on price but show a positive difference, most reliably domestic assault and weapons offenses, show a quantity response without a detectable price response. Theft and motor vehicle theft are null on both margins.

5.4 Rental Prices

I apply the same near–far design to lease listings, with the log asking rent as the outcome (Table 6). Three offenses lower the asking rent, but the estimates are sensitive to the fixed effects: burglary under parcel fixed effects, and sex offenses and liquor violations only under parcel-by-era fixed effects; all three survive the post-date placebo. Narcotics leaves rents unchanged under parcel fixed effects, where its post-date placebo is clean, though the placebo fails under parcel-by-era fixed effects; homicide is estimated too imprecisely in the lease sample to sign, in line with Klimova and Lee (2014), who find no effect of nearby murders on Sydney rents. Narcotics instead appears in the leasing frictions of Panel B: it raises the probability a listing returns to market (robust across all twenty-two specifications, with a null future-crime placebo; Figure OF3). The full offense battery is in Table OA27.

Table 6: The Rental Market

	PIN			PIN×era		
Offense	Min (%)	Max (%)	Sig. (plac.)	Min (%)	Max (%)	Sig. (plac.)
A. Asking rent (percent)						
Priced (asking rent lower; identifying FE in note)						
Burglary	−0.34*** (0.12)	−0.07 (0.06)	6/12 (0)	−0.30** (0.12)	−0.04 (0.07)	1/12 (0)
Sex offenses	−0.29* (0.15)	−0.21* (0.13)	1/12 (0)	−0.32 (0.27)	−0.16 (0.14)	7/12 (0)
Liquor law	−0.23 (0.26)	−0.01 (0.13)	0/12 (0)	−0.30*** (0.07)	0.09 (0.49)	4/12 (0)
Precise nulls (rent effect near zero, tightly estimated)						
Narcotics†	0.01 (0.06)	0.10 (0.08)	0/12 (0)	0.02 (0.17)	0.14 (0.09)	0/12 (4)
Vandalism	−0.06 (0.06)	0.05 (0.07)	0/12 (0)	−0.03 (0.04)	0.09 (0.07)	0/12 (1)
Prostitution	−0.16 (0.10)	0.00 (0.17)	0/12 (1)	−0.33*** (0.12)	−0.03 (0.14)	1/12 (0)
Imprecise / underpowered						
Homicide	−0.55 (0.61)	1.03 (1.20)	0/12 (0)	−0.59 (0.59)	1.16 (1.22)	0/12 (0)
B. Leasing frictions						
Narcotics, back-on-market (pp)	0.18** (0.08)	0.97*** (0.34)	11/12 (0)	0.27*** (0.08)	1.28*** (0.40)	11/12 (0)
Homicide, time on market (%)	1.01 (4.65)	12.19 (11.00)	0/12 (0)	1.54 (3.98)	19.63* (11.70)	4/12 (0)
Burglary, withdrawn (pp)†	−1.06** (0.46)	−0.02 (0.16)	3/12 (0)	−1.12** (0.53)	−0.08 (0.15)	5/12 (2)

Notes. Near–far estimates on lease listings, 2007–2025, tract × listing-month fixed effects, clustered by tract (118,849 listings with parcel fixed effects; 98,813 with parcel × era). For each specification, Min and Max are the most negative and most positive estimates across the twelve windows, followed by the number significant at 5%; in parentheses, windows with a significant post-listing placebo; †, placebo fails in two or more. Panel A: log asking rent. Panel B: return to market and withdrawal (percentage points), time on market (percent).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA28 shows how the listing agent's experience in the tract moderates the effect of nearby crime on rental listing outcomes. Near narcotics incidents, more experienced agents lease units faster, with days on market about 0.6 percent shorter per standard deviation of experience, and set slightly lower asking rents; the average rental discount stays small. The tenant's agent's experience is unrelated to asking rents or days on market in any window.

5.5 Heterogeneity

The point estimates differ across macroeconomic eras, but the differences are not statistically significant (Table 7). Burglary and vandalism discounts are larger in the 2009–2013 foreclosure crisis, homicide's in the 2014–2025 recovery, and narcotics is negative in each era (crisis −0.33, s.e. 0.23; recovery −0.52, s.e. 0.31). For no priced offense does a Wald test reject equal crisis and recovery discounts ($p = 0.41$ for homicide, 0.26 for burglary, 0.63 for narcotics, and 0.10 for vandalism). The crisis estimates for rare offenses rest on few within-crisis repeat sales and are imprecise, as their standard errors show.

Table 7: Crime Capitalization by Era

Offense	Identifying parcels	Pooled 2007–25	Crisis 2009–13	Recovery 2014–25
Priced				
Homicide	4,950	-3.02^{**}	-0.07	-4.12^{***}
		(1.38)	(4.73)	(1.47)
Narcotics	49,513	-0.29^{**}	-0.33	-0.52^{*}
		(0.14)	(0.23)	(0.31)
Burglary	76,323	-0.24^{**}	-0.49^{*}	-0.18
		(0.12)	(0.25)	(0.14)
Vandalism	88,765	-0.17^{**}	-0.54^{**}	-0.11
		(0.08)	(0.25)	(0.10)
Suggestive				
Disorderly conduct	19,550	-0.65^{**}	-0.87^{*}	-0.59^{*}
		(0.29)	(0.51)	(0.35)
Sex offenses	36,393	-0.47^{**}	-0.20	-0.36
		(0.23)	(0.72)	(0.23)
Precise nulls				
Larceny/theft	105,610	0.02	0.07	0.02^{**}
		(0.01)	(0.06)	(0.01)
Motor vehicle theft	70,464	0.09	0.00	0.08
		(0.12)	(0.38)	(0.12)
Assault (non-dom.)	89,931	-0.03	-0.07	-0.04
		(0.04)	(0.10)	(0.05)
Family (non-violent)	12,632	1.49^{**}	0.99	2.05^{**}
		(0.60)	(1.08)	(0.86)
Assault (domestic)	77,543	-0.03	0.18	-0.02
		(0.12)	(0.28)	(0.14)
Trespass	9,395	-0.09	-0.49	0.05
		(0.38)	(0.93)	(0.39)
Robbery	46,664	0.02	-0.58	0.06
		(0.18)	(0.45)	(0.19)
Liquor law	3,935	0.61	0.55	0.23
		(0.43)	(0.93)	(0.39)
Parcel FE		Yes	Yes	Yes
Tract × year-month FE		Yes	Yes	Yes
Crime × era interaction		No	Yes	Yes

Notes. δ as in Table 2. Offense groups as in Table 2: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Crisis (2009–2013) and recovery (2014–2025) effects come from an era-interacted model with 2007–2008 as the base; the pooled column is the full-sample effect and need not lie between them. Each offense is shown at its own central identifying window (the window differs across offenses, for example trespass at one month), held fixed across the three columns; identifying parcels (exposure varies within parcel) are counted at that window, whereas Table 2 counts them at twelve months. The crisis and recovery columns report per-era near–far levels, not deviations from the pooled column. Era differences are not formally tested.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The discounts also vary with neighborhood socioeconomic status. I use ten predetermined tract

characteristics: median household income, the poverty rate, the share with a college degree, the owner-occupancy rate, median home value, median gross rent, the Black and Hispanic population shares, racial segregation, and the vacancy rate. In separate regressions, I interact each standardized characteristic with the near–far term, controlling for the tract's predetermined crime level (Table OA29).

All four priced offenses carry larger discounts in lower-income and less-educated tracts; burglary and narcotics also carry larger discounts in higher-poverty tracts. A one-standard-deviation increase in tract income reduces the homicide discount by about 2.28 percentage points, with smaller reductions for burglary, narcotics, and vandalism. Discounts are likewise larger in tracts with lower median home values and rents, while neither owner-occupancy nor vacancy shows a consistent pattern. Because income, poverty, education, home values, and rents are strongly correlated, these results are best interpreted as a common socioeconomic gradient rather than separate findings. Table OA29 reports every offense and characteristic.

Discounts are also larger in tracts with a higher Black share. Controlling for predetermined tract crime, this gradient is marginally significant for homicide and burglary and significant for narcotics and vandalism. Hispanic share shows no significant gradient at the five-percent level for any priced offense (Table OA29). Because tracts with a higher Black share also tend to have lower incomes, I include income and racial composition jointly (Table OA30). With income and share Black together, both estimates lose precision for homicide, burglary, and narcotics, leaving their separate contributions unresolved. When income and racial segregation, measured by the divergence index (Roberto, 2016), are included jointly, the income gradient remains significant for burglary and narcotics, and at ten percent for homicide, while segregation shows no significant gradient. Vandalism is the exception: segregation remains significant (-0.56, s.e. 0.22), while income does not. This result may reflect a role for neighborhood stigma in the pricing of visible disorder, although the estimates do not establish that mechanism. Sampson and Raudenbush (2004) find that residents perceive more disorder in neighborhoods with more Black residents, holding observed disorder fixed.

5.6 Robustness

5.6.1 Threats to Identification

Reverse Causality and Mean Reversion Prices could drive crime rather than the reverse, or repeat-sales properties transacting at temporarily depressed prices could appreciate mechanically. I test both using a post-date placebo that replaces pre-transaction crime with crime occurring after the transaction, which cannot affect a price already set. The priced offenses pass this placebo at their central windows and in almost all specifications: post-sale crime is significant in none of the 63 specifications for homicide and vandalism, one for narcotics, and four for burglary (bottom panels of Figure 3 and Appendix Figures A1–A3). The exception is burglary, where post-sale crime is significant at two and six months (the six-month cases are the placebo hits in Appendix Figure A3); six months is also where burglary's pre-sale estimate is largest (-0.34, s.e. 0.14; Appendix

Table A4), so the placebo does not fully separate burglary's peak discount from post-sale crime.

Predetermined Neighborhood Controls The estimated discounts are not explained by pre-existing neighborhood investment or decline. I add five neighborhood controls, building permits (investment), vacant buildings (disinvestment), business licenses (commercial activity), tree canopy (amenity), and total 311 service requests (disorder), each entered as its own near–far difference alongside crime. Each varies across a parcel's sales and therefore retains identifying variation with parcel fixed effects. All five are predetermined relative to the twelve-month crime window: tree canopy is measured two years before the sale, while permits, vacant buildings, business licenses, and 311 requests are measured as cumulative stocks of events recorded more than twelve months before the close. Predetermined measures are used deliberately: contemporaneous neighborhood conditions may themselves respond to local crime, so conditioning on them could amount to controlling for an outcome of the exposure. These controls instead capture pre-existing neighborhood conditions rather than potential consequences of the crime itself (Appendix Table A7).

Point estimates change little (Appendix Table A7): a nearby homicide still lowers sale prices by about three percent, and the property-crime estimates move by hundredths of a percent. Precision falls for two offenses: with all five controls together, burglary (-0.22, s.e. 0.12) and narcotics (-0.26, s.e. 0.15) are significant only at ten percent. The priced effects therefore do not appear to reflect the measured physical or economic conditions of the block that are correlated with local crime.

Cross-Crime Confounding A discount attributed to one offense could in principle be capitalizing a spatially correlated neighbor. Homicide, burglary, and narcotics are near-orthogonal to the other offenses in the near ring, and entering all twenty-two offenses' near–far terms jointly, at a common twelve-month window, leaves each of their discounts nearly identical to its separate estimate (Online Appendix K; Table OA35). Vandalism is the exception: it correlates moderately with the high-volume nuisance offenses it co-occurs with, and its joint estimate is -0.02. At the common twelve-month window, however, vandalism's separate estimate is already small and insignificant (-0.05, s.e. 0.06), so the joint specification does not test its six-month discount, and part of that discount may reflect the co-occurring offenses. Disorderly conduct, by contrast, retains a discount in the joint specification (-0.58, significant at five percent).

Sample Selection The repeat-sales estimates use only properties that sell, so they would be biased if crime changed which properties sell. I test this in two ways. First, I regress the monthly probability of sale on nearby crime for all residential parcels, including those that never sell. Nearby crime does not change this probability: the estimate is statistically indistinguishable from zero ($p \approx 0.4$) and small relative to a monthly sale rate of about 0.3 percent. Crime does raise listings more than completed sales for several offenses (Table OA26), but it is completed sales that determine the repeat-sales sample. Second, a Heckman correction, which uses mortgage-rate lock-in to predict whether an owner sells, leaves the crime coefficients largely unchanged (Online Appendix L).

Distressed Sales Foreclosures and short sales could drive the discounts if distressed sellers cluster near crime. Allowing the near–far effect to differ for distressed sales (REO, short sale, or foreclosure), the discounts are if anything larger among distressed sales, but that subsample is thin near crime and the two differ significantly only for narcotics (Table OA31). Because distressed status itself lowers prices, this split shows where the discount appears rather than testing a forced-seller mechanism (Campbell et al., 2011; Ellen et al., 2013; Anenberg and Kung, 2014). A predetermined vacancy control, described above, leaves the discounts essentially unchanged.

Reporting Endogeneity Recorded incidents reflect both underlying crime and the chance that a crime is reported or detected, so the estimates could pick up reporting rather than crime if reporting rates move with prices. Two features limit this concern. The largest discount is for homicide, which is almost always recorded. And the near–far comparison within tract and month removes differences in reporting across neighborhoods and over time, so a remaining bias would require reporting to differ between a home's own block and blocks a few hundred yards away.

5.6.2 Exposure Specification and Inference

Selection into Zero Exposure Properties with no nearby crime could differ systematically from exposed properties along unobserved dimensions. I test for such selection using the mass-point dummy test of Caetano (2015); Caetano et al. (2021) for the three priced offenses with sufficient support at zero: narcotics, vandalism, and burglary. For narcotics and vandalism a large discontinuity in the raw data disappears under the headline parcel and tract×month fixed effects, and explicitly controlling for the boundary leaves the narcotics discount unchanged and, if anything, strengthens the vandalism discount. For burglary the boundary term is itself significant (-0.978, s.e. 0.317), so the linear discount is identified partly from the contrast between exposed and unexposed parcels. The discontinuity appears only once parcel fixed effects are included, so it does not reflect burglars targeting more valuable houses, a concern Gibbons (2004) raises for London. The jump does not reflect spillover from the adjacent ring: controlling for burglary at 0.05–0.10 miles leaves it unchanged (-0.99, s.e. 0.32). Because the test jointly detects failures of exogeneity, functional form, and parallel trends, its rejection does not establish the direction of any bias, and I treat the linear count for burglary as a maintained approximation; adding the boundary dummy moves δ from -0.244 (s.e. 0.117) to -0.552 (s.e. 0.168) (Online Appendix M, Figure OF13). At a 0–0.10 mile near ring, where 27 percent of sales have no nearby burglary and the boundary test does not reject (-0.06, s.e. 0.27), burglary's discount remains significant (-0.14, s.e. 0.07; Appendix Table A2), with no significant post-sale placebo at any window.

Functional Form How nearby crime enters the regression matters differently across offenses (Table OA34). Homicide, rare enough that a single incident is effectively the treatment, is significant whether exposure enters as a raw count, a presence indicator, or in inverse hyperbolic sine (-3.0 to -3.6 percent throughout). The property-crime discounts, by contrast, are significant at five percent

only under the raw count: narcotics is significant at ten percent under the presence indicator (−0.78, s.e. 0.43) and the inverse hyperbolic sine (−0.60, s.e. 0.31), and burglary and vandalism are not significant under either. A presence dummy weights one incident like four, and the inverse hyperbolic sine compresses high counts, so both dilute an effect that the dose-response (Table OA6) shows is driven by accumulated exposure rather than an isolated incident. I therefore adopt the raw count as a maintained approximation for the property offenses; homicide is significant under every form.

Across all twenty-two offenses (Table OA34), the choice of functional form does not manufacture the priced discounts; some offenses that do not price in the headline count specification do turn significant under the inverse-hyperbolic-sine or presence transforms, which rescale or redefine exposure. Two offenses outside the priced set are significant under all three forms: disorderly conduct, negative (−0.65, −1.00, and −0.81, the last at ten percent), and bad checks, positive (0.66, 1.06, and 1.07). Homicide is significant at five percent under every form and narcotics at ten percent under the presence and inverse-hyperbolic-sine forms; burglary and vandalism are significant under the raw count maintained above but not under the presence or inverse-hyperbolic-sine transforms.

Spatial Standard Errors The baseline standard errors are clustered by census tract (Cameron and Miller, 2015), allowing arbitrary correlation among parcels within a tract but not across tract boundaries. To allow for spatial correlation that crosses those boundaries, I also report Conley (1999) standard errors using a Bartlett kernel with cutoffs of 0.5 and 1.0 km. The 1.0 km cutoff exceeds twice the 0.30-mile outer radius used to construct crime exposure, so parcels with overlapping exposure areas can have correlated residuals. Tract clustering and Conley inference make different assumptions: the former allows unrestricted within-tract correlation regardless of distance, while the latter allows correlation across tracts but downweights it with distance and sets it to zero beyond the cutoff. This check reports both the era-salient and the pooled discount for each priced offense. Conley standard errors at 0.5 and 1.0 km, reported in Table OA32, are below the clustered standard errors for every priced offense; inference and the significance stars use the clustered ones. Clustering instead by Chicago's 77 community areas, a coarser grouping, leaves homicide and burglary significant at five percent ($p = 0.03$ and 0.04); vandalism is significant at ten percent ($p = 0.06$), and narcotics is not ($p = 0.12$) (Table OA32).

5.6.3 Specification Curves and Spatial Checks

Specification curves: sale prices Figure 3 reports the sale-price specification curve for vandalism (narcotics, homicide, and burglary appear in Appendix Figures A1–A3); each curve sorts the doorstep near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ across 63 specifications. The grid varies the exposure window from six to twelve months, the fixed-effect design (parcel + tract×month, or parcel×era + tract×month), the comparison ring (0.20–0.30, 0.25–0.35, 0.30–0.40 mile), the functional form (linear count, presence indicator, inverse hyperbolic sine), and the predetermined controls (none, all five jointly, or one at a time). The window begins at six months because shorter

windows fall inside the listing and under-contract period, where measured pre-sale exposure is sparse and partly contemporaneous with the sale itself. Nearly all specifications are negative for the four priced offenses; the exceptions are the presence-indicator estimates for burglary (0.54, s.e. 0.98) and vandalism (about 0.46), both imprecise. The discount is significant at five percent in 33 of 63 specifications for vandalism, 29 for narcotics, 26 for burglary, and 15 for homicide, whose point estimates range from about −3.6 percent to about zero with wide intervals. The post-sale placebo is significant in none of the 63 for homicide and vandalism, one for narcotics (the presence indicator at ten months), and four for burglary (the linear count at six months under each of the three far rings, and the inverse hyperbolic sine at eight months).[12] The full curve over all one-to-twelve-month exposure windows (103 specifications) is reported in Figures OF5–OF8.

Spatial Falsification A falsification test that moves the treatment band outward to 0.05–0.20 leaves narcotics, burglary, and vandalism significant in at most 1 of 48 specifications (Figure OF9), indicating effects concentrated at the doorstep. Homicide behaves differently. It is significant in 11 of 48 off-doorstep specifications, all in the 0.10–0.15 band, with a median estimate (−2.72) essentially identical to the doorstep (−2.78),[13] while the intervening 0.05–0.10 band is null. Homicide is therefore robust at the doorstep but, unlike the other three offenses, not localized to it. This is consistent with Section 5.2, where homicide is priced through public incidents rather than the dwelling, a pattern that is neighborhood-scale rather than confined to the doorstep. Because homicide reaches beyond the doorstep, it likely raises exposure in the 0.20–0.30 mile comparison ring as well; since $\delta = \beta_{\text{near}} - \beta_{\text{far}}$ nets that ring out, the reported homicide discount is if anything a lower bound on the doorstep effect rather than an overstatement. Significantly negative placebo estimates off the doorstep (3 of 48 for narcotics, 5 of 48 for burglary and vandalism) modestly exceed the 5 percent expected by chance and are concentrated in the wider bands, consistent with the contamination this falsification test is designed to detect rather than with a doorstep effect.

Specification curves: rents The rental specification curves cover twenty-two specifications: eleven exposure windows (two to twelve months) under each of two fixed-effect designs (parcel, and parcel by era). The narcotics estimate on the probability a lease listing returns to the market is positive and significant in all twenty-two, with no significant placebo (Figure OF3). The sex-offense estimate on the asking rent is significant in eight of the same twenty-two specifications, with no significant placebo (Figure OF4).

[12] The specification curve shows how stable the estimates are across modeling choices. Because the specifications share data and most choices, they are not independent, so the number of significant estimates is not a formal test.

[13] The −2.78 and −2.72 medians are taken over the significant estimates in the band-varying close specification grid (window × fixed effects × far ring), the object common to both bands; the larger doorstep significant count (15) additionally varies the predetermined controls and the functional form.

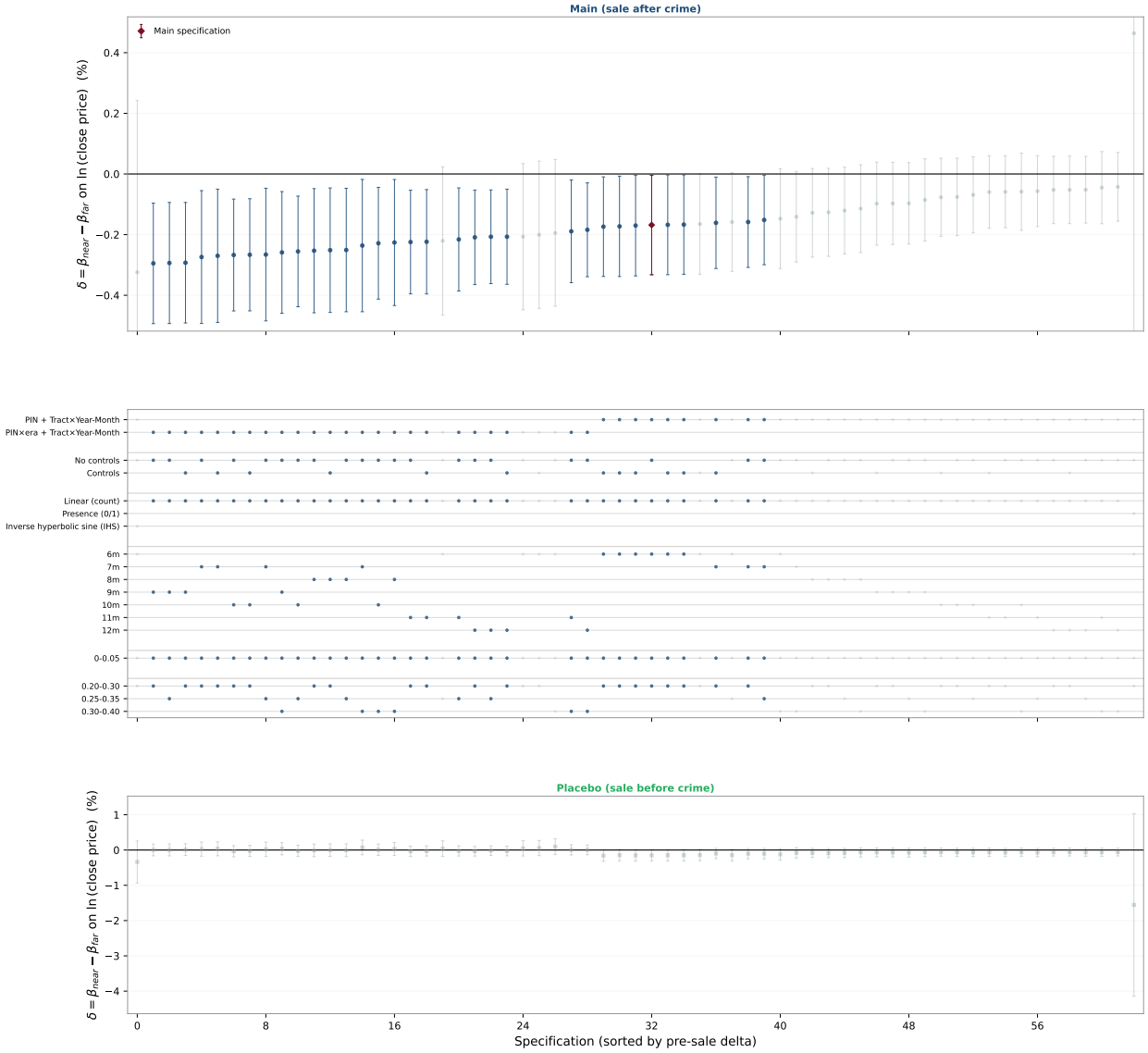


Figure 3: Vandalism: Sale-Price Discount Across Specifications

Notes. Common to this figure and Appendix Figures A1–A3; specifications as described in Section 5.6. Top panel: δ (%) with 95 percent intervals, sorted from most to least negative; navy marks estimates significant and negative at five percent, and maroon marks the main specification (linear count, parcel + tract×month fixed effects, 0.20–0.30 mile comparison ring, no controls) at each offense's central window: ten months for homicide and narcotics, eight for burglary, and six for vandalism. Middle panel: specification choices. Bottom panel: the post-sale placebo, which replaces pre-sale crime with crime after the sale. All years pooled (2007–2025); standard errors clustered by tract. Vandalism: 33/63 significant at five percent; placebo 0/63.

6 Discussion

6.1 Why most offenses are not priced

Neither frequency nor severity alone explains which offenses are priced. Homicide is rare and carries the largest discount, while larceny, motor vehicle theft, and non-domestic assault rule out discounts as large as burglary's. The distinction between risks anchored to an address and those that travel with the household offers one interpretation (Section 2, Table OA19). All four priced offenses are anchored to the location; among eleven offenses that travel with the household, only sex offenses show a suggestive discount (Fisher's exact test, $p = 0.09$). This supports a role for relevance, although the groups also differ in severity, frequency, and visibility.

Measured predictability does not explain the pattern either. Crime on surrounding blocks predicts subsequent crime near the property but carries no detectable discount, and the most predictive offenses are not those that are priced (Section 5.2). These findings are consistent with selective attention to crime information (Bordalo et al., 2012; Gabaix, 2014), although recorded-crime forecasts need not match buyers' expectations. Within burglary, frontage incidents lower prices while interior incidents have no detectable effect, with a significant difference at the longest exposure windows (Table 3). This supports a role for observability and may help explain the contrast with Gibbons (2004), whose burglary measure pools incidents within 250 meters across locations that may differ in visibility.

The evidence therefore points to relevance and observability rather than predictive content, without separately identifying their contributions. An undetected discount may reflect limited awareness, little relevance to the property, unchanged expectations, or limited precision. Changes in crime information displayed by real estate platforms, discussed in Section 7, would provide a more direct test of the information channel.

6.2 Why asking prices respond little

The absence of a detectable average asking-price response to homicide, alongside a roughly three-percent closing-price discount, is consistent with the framework's fixed seller reservation value, although asking prices do not directly measure it (Table 4). Two explanations may account for the muted response. Sellers may anchor on their purchase price or a previous valuation: Genesove and Mayer (2001) show that owners facing prospective nominal losses set higher asking prices and sell more slowly. Alternatively, comparative market analyses may rely on nearby sales that predate the incident. The more negative asking-price responses among experienced listing agents are consistent with information or expertise beyond these comparables, but the data do not distinguish between the explanations.

Closing-price discounts could also reflect changes in sellers' bargaining positions. For burglary, narcotics, and vandalism, discounts do not differ detectably between soft and tight markets ($p > 0.4$); homicide estimates are too imprecise for an informative comparison. These results do not rule out bargaining effects, and the contract-timing and distressed-sale tests in Online Appendix N also

have limited power.

6.3 Who carries the information

The agent results distinguish local knowledge from general experience. For homicide, the buyer-agent gradient in closing prices remains after controlling for total transactions, while total experience has no detectable gradient. The listing-agent gradient survives this control at only one of twelve windows. For narcotics, the asking-price gradient instead reflects listing agents' total experience, suggesting general pricing expertise. These patterns relate to evidence that agents' information advantages affect transaction outcomes (Levitt and Syverson, 2008), but nonrandom agent assignment prevents a causal interpretation of the gradients.

News coverage may also contribute to information reaching buyers. Homicide receives roughly fifteen times as many stories per incident as the next-highest offense in Media Cloud[14] and the largest article volume in GDELT,[15] and it carries the largest discount. Yet category-level coverage does not measure whether buyers learn of a particular incident, and narcotics is priced despite far less coverage. Klimova and Lee (2014) also find no additional house-price effect from more heavily covered murders in Sydney. These comparisons leave the relative importance of media, agents, and local contacts unresolved.

Neighborhood reputation may influence how buyers interpret an incident. Vandalism discounts are larger in more segregated tracts even after controlling for income (Table OA30). This parallels Sampson and Raudenbush (2004), who find greater perceived disorder in neighborhoods with more Black residents, holding observed disorder constant. The result is consistent with stigma amplifying the response to visible disorder, but does not establish that mechanism.

6.4 Welfare and policy

The estimates measure price responses among completed transactions, not average household willingness to pay to avoid crime. Buyers most averse to crime may withdraw from consideration, leaving a selected group of purchasers. Negotiated prices also reflect seller valuations and bargaining positions, while buyers may consider resale consequences as well as current occupancy. The transaction data do not separate these influences.

Evaluations of crime-reduction policies based only on sale prices may therefore omit important benefits. Longer marketing periods can entail carrying costs and search effort; withdrawals may represent forgone moves, postponement, or alternative uses of the property. Their welfare implications depend on what follows. Nor do price effects rank offenses by victim costs (McCollister et al., 2010). A small or undetected discount need not imply a small benefit from preventing an offense, particularly when buyers rarely learn of it.

[14] Media Cloud, an open-source platform for large-scale quantitative news analysis (`https://mediacloud.org`); the measure is the count of news articles mentioning an offense category over the sample period, divided by the number of recorded incidents in that category.

[15] The GDELT Project, a global open database that monitors and indexes news coverage worldwide (`https://www.gdeltproject.org`).

Information disclosure raises a separate policy question. If observability contributes to price responses, crime maps or disclosure rules may affect property values without changing underlying crime. A resulting price decline need not indicate a welfare loss: buyers may simply be better informed about an existing risk. The analysis motivates this distinction but does not estimate the effects of disclosure on prices or welfare.

6.5 Scope

For offenses recorded through enforcement, price responses may reflect policing as well as underlying crime. Enforcement affects recorded offense counts (Levitt, 1998) and can itself affect local well-being (Ang, 2021; Legewie and Fagan, 2019). If buyers view concentrated drug or vice enforcement as a disamenity, the design cannot separate that response from the response to crime. Among the four priced offenses, this concern applies most directly to narcotics.

The estimates summarize the price effect per additional incident within 0.05 miles. The distance-based design can be applied elsewhere, although the same radius may encompass different numbers of properties and blocks. Whether the magnitudes generalize depends on local density, street layout, housing-market conditions, and how buyers learn about nearby incidents. Responses also vary with neighborhood income within Chicago (Section 5.5), making the local income distribution relevant to comparisons across settings.

The rental and sale estimates differ for several reasons. The smaller lease sample generally yields less precise estimates and consists of formal, MLS-listed units concentrated in condominiums (Table A1). Differences in property composition may therefore contribute to differences in treatment effects. Renters and buyers also differ demographically and may value safety differently. A lease buys occupancy, whereas a purchase also exposes the owner to future buyers' assessments through resale. Prices also form differently: asking rents reflect landlords' pricing decisions and anticipated demand, while sale prices reflect negotiation between buyers and sellers. These differences prevent attributing the sale–rent gap solely to resale exposure. The rental findings describe posted rents in the formal MLS market and need not generalize to informal or lower-income rentals.

7 Conclusion

Only about one in five prospective homebuyers is willing to compromise on personal safety to afford a home (Anderson, 2025), yet most types of crime do not measurably lower house prices. Using repeat sales of the same property and comparing crime on its block with crime a few blocks away in the same neighborhood and month, I find sale-price discounts for only four of twenty-two offense categories. Neither severity nor measured recurrence alone explains which offenses are priced.

Three findings help interpret this pattern. First, price effects are more localized than the predictive relationships in the crime data. Crime a few blocks away predicts subsequent crime near the property, yet detectable price effects are confined to the immediate block for burglary and narcotics and extend only a few blocks for homicide. Second, price effects vary with where an

incident occurs. Frontage burglaries, which are more exposed to view, lower prices, while burglaries inside the dwelling have no detectable effect. The difference grows with the exposure window and is significant at twelve months, consistent with a role for observability. Third, discounts emerge in closing prices rather than asking prices, and a nearby homicide also lengthens time on market and raises the probability of withdrawal. Asking prices respond more negatively to some nearby crimes when listing agents are more experienced.

These patterns are consistent with price responses depending on whether buyers learn of an incident, regard it as relevant to the property, and revise their expectations of future exposure. An absent discount may therefore reflect limited awareness, no revision in expected exposure, or an effect too small to detect. Under the proposed framework, observed discounts need not equal willingness to pay to avoid crime. Property values need not rank offenses by the harm they impose, and price-based evaluations may understate the benefits of reducing crime that buyers rarely observe.

Future research could test how access to crime information affects the price effects of crime. The introduction and subsequent removal of crime maps from online real estate platforms offer one setting for such a test.[16] If access to these maps informs buyers about nearby incidents, discounts should increase when the information becomes available and diminish when it is removed, holding underlying crime fixed. The change should be largest for offenses displayed on the maps. The framework could also guide research on other local risks, such as flooding, by distinguishing changes in underlying risk from changes in what buyers know about it.

[16] Zhang et al. (2022) propose a related design based on the introduction of public crime maps.

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A Appendix

This appendix reports seven supporting tables and the sale-price specification curves for narcotics, homicide, and burglary. Unless a note says otherwise, each reports $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$, the percent sale-price change per incident within 0–0.05 miles of the property against 0.20–0.30 miles, with parcel and tract × year-month fixed effects and standard errors clustered by census tract; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All other supporting material is in the Online Appendix.

Table A1: Summary Statistics

	Mean	SD	p10	Median	p90	Obs.
A. Sale listings ($N = 530,395$)						
Close price ($)	366,777	382,539	74,000	279,000	700,000	530,395
Log close price	12.44	0.92	11.21	12.54	13.46	530,395
Original list price ($)	384,828	410,804	85,000	289,900	725,000	529,956
Days on market	96	147	5	42	248	529,995
Any list-price cut	0.68	0.47	0.00	1.00	1.00	530,395
Bedrooms	3.0	1.7	1.0	3.0	5.0	517,982
Bathrooms	2.1	1.1	1.0	2.0	3.0	530,395
Living area (sq. ft.)	1,175	1,115	0	1,100	2,483	454,018
Total rooms	7.1	3.2	4.0	6.0	11.0	517,753
Year built	1954	41	1902	1954	2007	453,343
Garage spaces	1.6	0.6	1.0	2.0	2.0	351,747
Fireplaces	0.9	0.7	0.0	1.0	2.0	206,694
B. Lease listings ($N = 191,155$)						
Monthly rent ($)	2,208	1,086	1,250	1,950	3,400	191,155
Log monthly rent	7.61	0.41	7.13	7.58	8.13	191,155
Days on market	43	57	6	25	102	191,122
Bedrooms	1.8	1.0	1.0	2.0	3.0	191,031
Bathrooms	1.6	0.8	1.0	1.0	2.0	191,155
Living area (sq. ft.)	883	675	0	900	1,691	191,041
Total rooms	4.7	1.4	3.0	5.0	6.0	191,032
Year built	1972	37	1912	1982	2008	150,337

Notes: Chicago MLS single-family, condominium, and residential-income listings, 2007–2025, on parcels with a constant parcel count. Panel A: closed sales ($10,000–$10,000,000). Panel B: leases (monthly rent $500–$10,000). Any list-price cut: at least one reduction before sale.

Table A2: The Near–Far Discount Across Nine Ring Definitions

Near vs. far ring (miles)	Homicide	Burglary	Narcotics	Vandalism
0–0.05 vs. 0.20–0.30 *(headline)*	-3.02 (1.38)	-0.24 (0.12)	-0.29 (0.14)	-0.17 (0.08)
0–0.05 vs. 0.25–0.35	-2.75 (1.39)	-0.24 (0.12)	-0.32 (0.14)	-0.16 (0.08)
0–0.05 vs. 0.30–0.40	-2.66 (1.35)	-0.27 (0.12)	-0.32 (0.14)	-0.15 (0.08)
0–0.10 vs. 0.20–0.30	-0.42 (0.67)	-0.14 (0.07)	0.01 (0.06)	-0.06 (0.05)
0–0.15 vs. 0.20–0.30	-0.90 (0.54)	-0.05 (0.05)	0.05 (0.04)	0.01 (0.03)
0–0.20 vs. 0.20–0.30	-0.74 (0.46)	-0.03 (0.04)	0.02 (0.03)	-0.02 (0.03)
0.05–0.10 vs. 0.20–0.30	0.55 (0.78)	-0.09 (0.07)	0.06 (0.07)	-0.02 (0.05)
0.10–0.15 vs. 0.20–0.30	-1.16 (0.64)	0.04 (0.06)	0.06 (0.05)	0.07 (0.04)
0.15–0.20 vs. 0.20–0.30	-0.45 (0.58)	-0.00 (0.05)	-0.02 (0.04)	-0.05 (0.04)

Notes: Each row is one near/far ring pair: the headline (row 1), the far ring moved out (rows 2–3), the near disk widened (rows 4–6), and an outer annulus in place of the doorstep (rows 7–9). Windows: homicide 10, burglary 8, narcotics 10, vandalism 6 months. $N = 242{,}815$.

Table A3: Sale-Price Capitalization, All Twenty-Two Estimable Offenses

Offense	Identifying parcels	Sig. discount windows	Min δ (s.e.) [mo]	Max δ (s.e.) [mo]
Priced				
Homicide	5,701	2/12	-5.86^{*} (3.35) [1]	-1.24 (1.43) [8]
Narcotics	53,844	5/12	-0.93 (0.75) [1]	-0.18 (0.19) [6]
Burglary	88,094	6/12	-0.34^{**} (0.14) [6]	-0.18^{*} (0.09) [12]
Vandalism	104,661	2/12	-0.17^{**} (0.08) [6]	0.26 (0.22) [1]
Suggestive				
Disorderly conduct	19,550	1/12	-1.33^{*} (0.78) [2]	-0.46 (0.54) [4]
Sex offenses	36,393	1/12	-0.47^{**} (0.23) [12]	0.11 (0.35) [3]
Precise nulls				
Theft	115,548	0/12	0.01 (0.02) [12]	0.08^{***} (0.03) [1]
Motor vehicle theft	85,781	0/12	-0.02 (0.09) [12]	0.51^{*} (0.30) [1]
Assault (non-dom.)	91,783	0/12	-0.05 (0.06) [7]	0.01 (0.11) [3]
Bad checks	8,919	0/12	0.66^{*} (0.35) [11]	2.20^{**} (1.03) [1]
Family (nonviol.)	15,928	0/12	1.00^{*} (0.51) [12]	2.16^{***} (0.82) [5]
Fraud	67,030	0/12	-0.36 (0.23) [1]	-0.04 (0.09) [6]
Assault (domestic)	90,500	0/12	-0.19^{*} (0.10) [12]	0.02 (0.15) [5]
Trespass	54,025	0/12	-0.15 (0.14) [6]	0.18 (0.22) [3]
Robbery	65,876	0/12	-0.06 (0.14) [9]	0.30 (0.42) [1]
Forgery	23,614	0/12	-0.07 (0.59) [1]	0.47 (0.45) [2]
Liquor	6,393	0/12	0.08 (1.20) [1]	1.34 (0.91) [2]
Prostitution	6,326	0/12	-1.00 (0.82) [1]	0.08 (0.12) [12]
Inconclusive				
Weapons	32,294	0/12	-0.62 (0.47) [5]	0.11 (0.58) [3]
Arson	6,116	0/12	-0.20 (1.06) [11]	1.27 (3.58) [1]
Gambling	3,125	0/12	-4.77 (4.33) [3]	1.99 (8.32) [1]
Stolen property	1,262	0/12	-2.23 (2.63) [2]	0.81 (1.20) [8]

Notes. All twenty-two estimable offenses under the specification of Table 2; kidnapping, embezzlement, extortion, child pornography, and other offenses have too few incidents to estimate and are excluded; columns as there, with Min δ and Max δ the most negative and most positive estimates across the twelve windows; offense groups as in Table 2: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Sig. discount windows counts windows with a significantly negative estimate at five percent, so positive significant estimates (bad checks, family offenses) count as zero.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Sale-Price Capitalization at All Twelve Exposure Windows

	Exposure window (cumulative months)											
Offense	1	2	3	4	5	6	7	8	9	10	11	12
Priced												
Homicide	−5.86*	−2.23	−1.45	−2.04	−1.28	−1.30	−1.59	−1.24	−2.54*	−3.02**	−3.15**	−2.49*
	(3.35)	(2.60)	(2.46)	(2.05)	(1.82)	(1.77)	(1.53)	(1.43)	(1.36)	(1.38)	(1.38)	(1.36)
Narcotics	−0.93	−0.37	−0.35	−0.32	−0.28	−0.18	−0.20	−0.38**	−0.31**	−0.29**	−0.35***	−0.33***
	(0.75)	(0.42)	(0.38)	(0.27)	(0.23)	(0.19)	(0.17)	(0.18)	(0.15)	(0.14)	(0.13)	(0.11)
Burglary	−0.20	−0.23	−0.20	−0.21	−0.25*	−0.34**	−0.30**	−0.24**	−0.22**	−0.22**	−0.22**	−0.18*
	(0.31)	(0.20)	(0.17)	(0.16)	(0.15)	(0.14)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.09)
Vandalism	0.26	0.13	−0.06	−0.11	−0.16*	−0.17**	−0.16**	−0.13*	−0.10	−0.08	−0.06	−0.05
	(0.22)	(0.15)	(0.12)	(0.10)	(0.09)	(0.08)	(0.08)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)
Suggestive												
Disorderly conduct	−1.22	−1.33*	−0.65	−0.46	−0.71	−0.66	−0.58	−0.50	−0.63*	−0.60*	−0.57*	−0.65**
	(1.03)	(0.78)	(0.63)	(0.54)	(0.48)	(0.44)	(0.38)	(0.35)	(0.33)	(0.32)	(0.29)	(0.29)
Sex offenses	−0.08	0.10	0.11	−0.02	−0.29	−0.31	−0.25	−0.20	−0.26	−0.39	−0.43*	−0.47**
	(0.64)	(0.42)	(0.35)	(0.32)	(0.31)	(0.27)	(0.26)	(0.25)	(0.25)	(0.25)	(0.24)	(0.23)
Precise nulls												
Larceny/theft	0.08***	0.04*	0.04*	0.03**	0.03*	0.02	0.02	0.02	0.01	0.02	0.01	0.01
	(0.03)	(0.02)	(0.02)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)	(0.02)	(0.01)	(0.02)	(0.02)
Motor vehicle theft	0.51*	0.28	0.17	0.10	0.05	0.04	0.09	0.09	0.05	0.06	0.07	−0.02
	(0.30)	(0.22)	(0.18)	(0.16)	(0.14)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.09)	(0.09)
Assault (non-dom.)	−0.05	−0.02	0.01	−0.03	−0.04	−0.04	−0.05	−0.03	−0.02	−0.01	−0.03	−0.04
	(0.20)	(0.14)	(0.11)	(0.09)	(0.08)	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)	(0.04)	(0.04)
Bad checks	2.20**	2.00***	1.36**	1.38**	1.09**	1.18***	1.08**	0.97**	0.77*	0.72**	0.66*	0.66**
	(1.03)	(0.74)	(0.61)	(0.57)	(0.53)	(0.45)	(0.43)	(0.41)	(0.39)	(0.37)	(0.35)	(0.33)
Family (non-violent)	1.99	1.07	1.16	1.68*	2.16***	1.77**	1.89***	1.68***	1.49**	1.28**	1.07*	1.00*
	(1.87)	(1.33)	(1.09)	(0.92)	(0.82)	(0.75)	(0.66)	(0.62)	(0.60)	(0.57)	(0.55)	(0.51)
Fraud	−0.36	−0.29*	−0.11	−0.07	−0.06	−0.04	−0.05	−0.05	−0.05	−0.08	−0.11	−0.11
	(0.23)	(0.17)	(0.12)	(0.10)	(0.10)	(0.09)	(0.10)	(0.09)	(0.09)	(0.08)	(0.08)	(0.08)
Assault (domestic)	0.02	0.00	−0.02	0.02	0.02	0.01	−0.03	−0.09	−0.17	−0.13	−0.18*	−0.19*
	(0.37)	(0.26)	(0.21)	(0.17)	(0.15)	(0.13)	(0.12)	(0.12)	(0.11)	(0.10)	(0.10)	(0.10)
Trespass	−0.09	0.16	0.18	0.00	−0.10	−0.15	−0.14	−0.12	−0.06	−0.05	−0.09	−0.14
	(0.38)	(0.29)	(0.22)	(0.18)	(0.15)	(0.14)	(0.13)	(0.12)	(0.11)	(0.10)	(0.10)	(0.10)
Robbery	0.30	0.15	0.09	0.08	0.04	0.02	−0.02	−0.03	−0.06	−0.01	0.01	−0.03
	(0.42)	(0.30)	(0.24)	(0.22)	(0.19)	(0.18)	(0.16)	(0.15)	(0.14)	(0.13)	(0.13)	(0.13)
Forgery	−0.07	0.47	0.12	0.15	0.05	0.38	0.44*	0.39*	0.30	0.30*	0.23	0.21
	(0.59)	(0.45)	(0.38)	(0.32)	(0.28)	(0.24)	(0.22)	(0.20)	(0.19)	(0.18)	(0.18)	(0.17)
Liquor law	0.08	1.34	0.71	0.72	0.47	0.39	0.61	0.65	0.64*	0.51	0.36	0.36
	(1.20)	(0.91)	(0.81)	(0.60)	(0.52)	(0.48)	(0.43)	(0.40)	(0.38)	(0.36)	(0.35)	(0.35)
Prostitution	−1.00	−0.43	−0.23	−0.23	−0.18	−0.28	−0.14	−0.04	0.01	0.01	0.05	0.08
	(0.82)	(0.51)	(0.48)	(0.39)	(0.35)	(0.33)	(0.30)	(0.27)	(0.22)	(0.18)	(0.15)	(0.12)
Inconclusive												
Weapons	−0.47	0.07	0.11	−0.51	−0.62	−0.39	−0.18	−0.12	−0.17	0.04	0.09	0.10
	(0.92)	(0.71)	(0.58)	(0.48)	(0.47)	(0.44)	(0.40)	(0.37)	(0.35)	(0.31)	(0.30)	(0.28)
Arson	1.27	0.32	0.27	0.10	−0.13	0.07	1.07	0.57	−0.15	−0.13	−0.20	0.07
	(3.58)	(2.75)	(2.33)	(2.01)	(1.78)	(1.63)	(1.49)	(1.31)	(1.22)	(1.15)	(1.06)	(1.03)
Gambling	1.99	1.69	−4.77	−1.39	−1.35	−0.76	−1.17	−3.04	−2.84	−2.91*	−1.79	−1.94
	(8.32)	(5.78)	(4.33)	(3.99)	(3.39)	(3.12)	(2.74)	(2.43)	(2.10)	(1.75)	(1.77)	(1.75)
Stolen property	−2.23	−2.23	−1.78	−2.02	−1.14	−0.67	0.26	0.81	0.08	0.19	−0.25	−0.43
	(4.06)	(2.63)	(2.18)	(1.95)	(1.83)	(1.75)	(1.59)	(1.20)	(1.06)	(1.02)	(1.05)	(1.05)

Notes. δ at every cumulative window from one to twelve months, under the specification of Table 2; offense groups as there: *priced*, a significant discount at two or more windows; *suggestive*, a significant discount at only one window; *precise nulls*, no significant discount and intervals that rule out a one-percent discount in at least nine of twelve windows; *inconclusive*, no significant discount but intervals too wide to rule out a one-percent discount. Covers the twenty-two estimable categories, including the white-collar offenses, prostitution, gambling, and stolen property. Standard errors in parentheses.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5: Rational Forecast: The Discount Does Not Track Forecastable Risk

	0–6 months		0–12 months	
Offense	Joint F	Partial R^2	Joint F	Partial R^2
Homicide†	127.3^{***}	0.004	237.0^{***}	0.009
Burglary†	1580.8^{***}	0.047	1871.1^{***}	0.062
Narcotics†	411.1^{***}	0.038	251.5^{***}	0.037
Vandalism†	711.6^{***}	0.045	627.5^{***}	0.054
Theft	6.8^{***}	0.035	9.1^{***}	0.017
Motor-vehicle theft	789.0^{***}	0.047	721.1^{***}	0.052
Robbery	357.2^{***}	0.045	365.8^{***}	0.060
Assault (non-dom.)	103.6^{***}	0.030	99.3^{***}	0.025
Assault (domestic)	696.5^{***}	0.062	497.1^{***}	0.058
Arson	80.3^{***}	0.001	204.3^{***}	0.002
Bad checks	27.4^{***}	0.002	34.0^{***}	0.005
Disorderly conduct	129.5^{***}	0.012	132.0^{***}	0.020
Family offenses	109.4^{***}	0.002	148.8^{***}	0.004
Forgery	82.3^{***}	0.012	91.1^{***}	0.018
Fraud	102.1^{***}	0.087	75.4^{***}	0.075
Gambling	170.2^{***}	0.013	261.6^{***}	0.029
Liquor	30.3^{***}	0.001	34.5^{***}	0.003
Prostitution	18.0^{***}	0.007	10.9^{***}	0.007
Sex offenses	75.8^{***}	0.008	45.2^{***}	0.014
Stolen property	3.2^{*}	0.000	5.8^{**}	0.000
Trespass	22.4^{***}	0.019	20.2^{***}	0.022
Weapons	251.0^{***}	0.054	348.9^{***}	0.067

Notes. Estimated without fixed effects. Outcome: crime within 0.05 miles after the sale. Predictor: crime 0.20–0.30 miles away before the sale, controlling for the doorstep's own prior crime. Windows: 0–6 and 0–12 months. F tests that the far-ring term is zero; partial R^2 is the share of the remaining variation the far ring explains. A predictive benchmark, not a causal estimate.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. † priced offense.

Table A6: Agent Experience Gradient on Price

Offense	Discount (mean agent)	Experience gradient (per SD) no agent FE	agent FE	Sig. windows (noFE/FE)
A. Asking price (listing agent)				
Narcotics	−0.01 (0.04)	-0.15^{***}	-0.09^{**}	9/7
		(0.05)	(0.04)	
Vandalism	0.02 (0.03)	-0.11^{***}	-0.07^{**}	8/3
		(0.03)	(0.03)	
Homicide	0.18 (0.59)	−0.82	-1.01^{**}	0/7
		(0.63)	(0.49)	
Burglary	−0.05 (0.10)	−0.04	0.02	0/0
		(0.11)	(0.08)	
B. Closing price (buyer's agent)				
Homicide	−0.94 (0.79)	−0.01	-1.93^{**}	0/6
		(1.70)	(0.89)	
Narcotics	−0.09 (0.12)	-0.52^{**}	0.13	3/0
		(0.25)	(0.15)	
Vandalism	−0.13 (0.11)	−0.04	0.10	0/0
		(0.17)	(0.15)	
Burglary	−0.07 (0.08)	0.25^{*}	0.12	0/0
		(0.15)	(0.13)	

Notes. Gradient per standard deviation of agent experience ($E = \ln(1 +$ prior same-agent same-tract listings), s.d. 1.20), each offense at its central window. Panel A: listing agent, asking price. Panel B: buyer's agent, closing price. Discount: near–far effect at mean experience. Gradients are estimated without and with agent fixed effects; Sig. windows counts significant windows of twelve, without/with. Sample: listings with an agent identifier.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A7: Priced Discounts with Predetermined Neighborhood Controls

	Homicide	Burglary	Narcotics	Vandalism
δ, baseline (no neighborhood control)	-3.02^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
Adding each predetermined control alone (near & far, stock to 12 months before close):				
+ Building permits (investment)	-3.03^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
+ Vacant buildings (disinvestment)	-3.06^{**}	-0.23^{**}	-0.25^{*}	-0.17^{*}
	(1.40)	(0.12)	(0.15)	(0.08)
+ Business licenses (commercial)	-2.95^{**}	-0.24^{**}	-0.30^{**}	-0.17^{**}
	(1.39)	(0.12)	(0.14)	(0.08)
+ Tree canopy (amenity)	-3.02^{**}	-0.24^{**}	-0.29^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
+ 311 service requests, all types (disorder)	-3.00^{**}	-0.24^{**}	-0.30^{**}	-0.17^{**}
	(1.38)	(0.12)	(0.14)	(0.08)
δ, all five controls together	-2.97^{**}	-0.22^{*}	-0.26^{*}	-0.17^{**}
	(1.40)	(0.12)	(0.15)	(0.08)

Notes. δ at each offense's central window. Row 1 is the baseline; the middle rows add one neighborhood control at a time; the last row adds all five. Each control is a near–far cumulative stock of events recorded more than twelve months before the close (tree canopy, two years before the sale); 311 data begin in 2011 and change composition in 2019 (Section 3).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

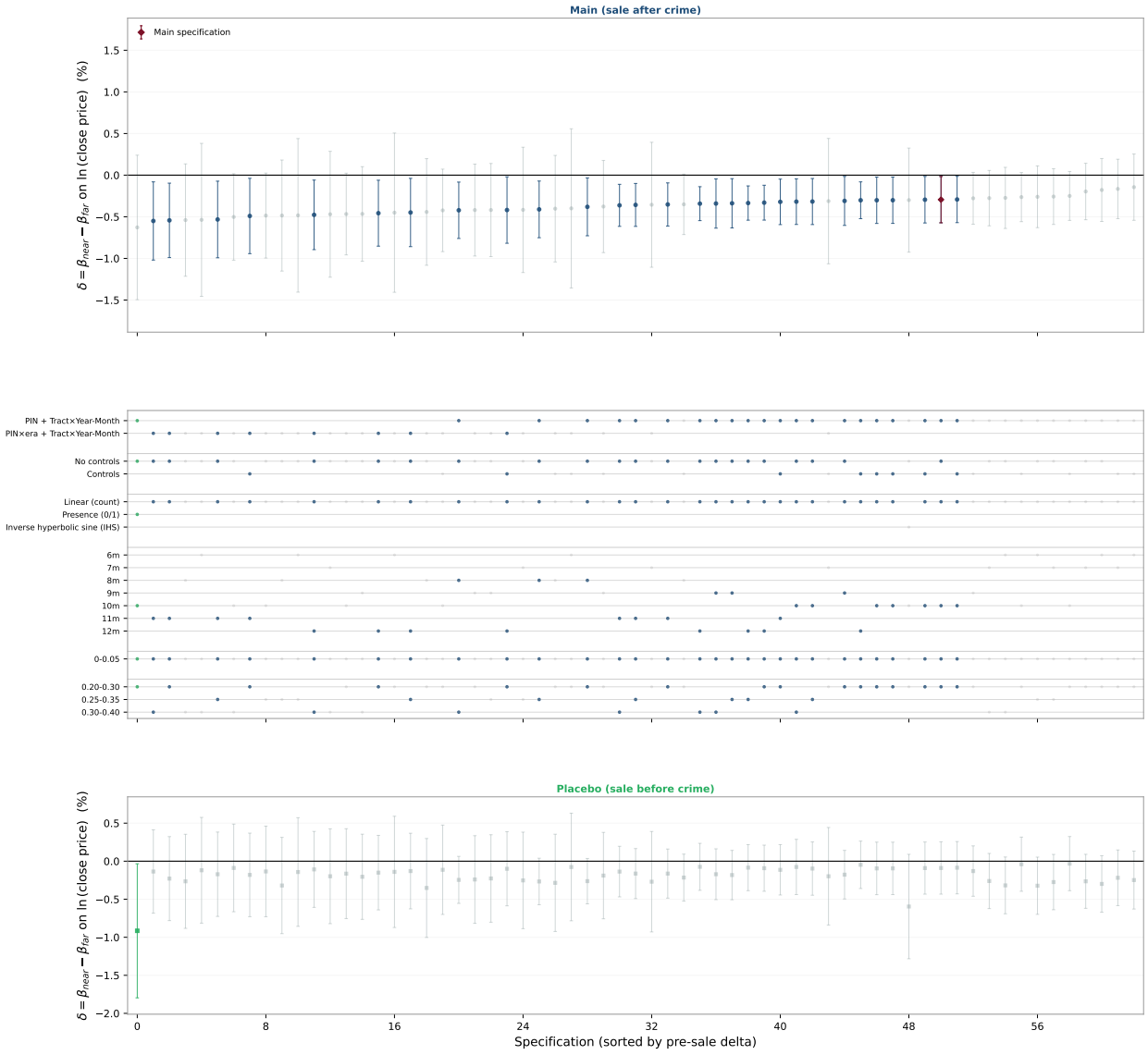


Figure A1: Narcotics: Sale-Price Discount Across Specifications

Notes. As in Figure 3, for narcotics. Main: 29/63 significant at five percent; placebo 1/63 (the presence indicator at ten months).

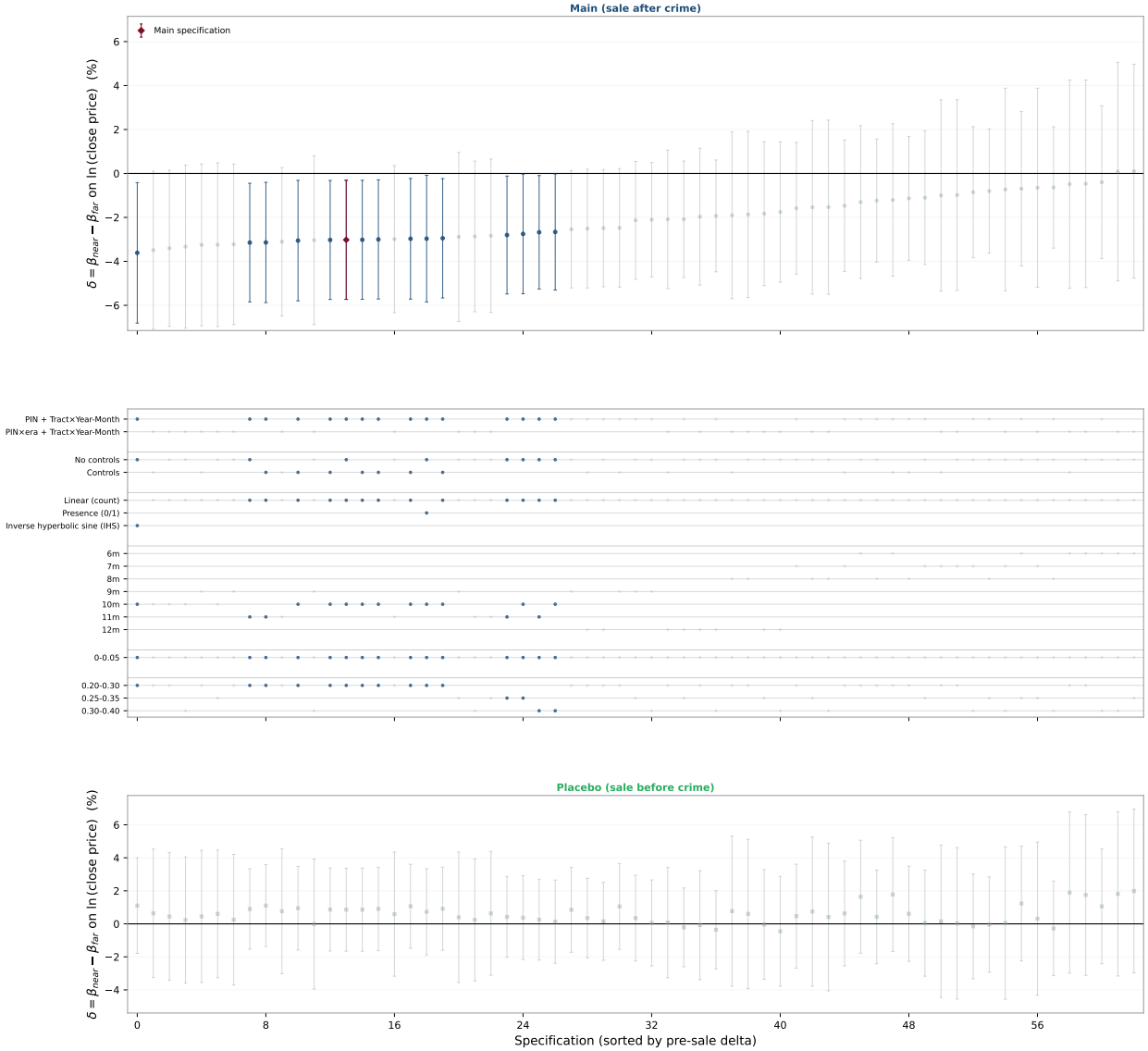


Figure A2: Homicide: Sale-Price Discount Across Specifications

Notes. As in Figure 3, for homicide. Main: 15/63 significant at 5%; placebo: 0/63. Homicide's intervals are an order of magnitude wider because incidents are rare, and the significant specifications cluster at the ten-to-eleven-month window; point estimates range from about −3.6 percent to about zero.

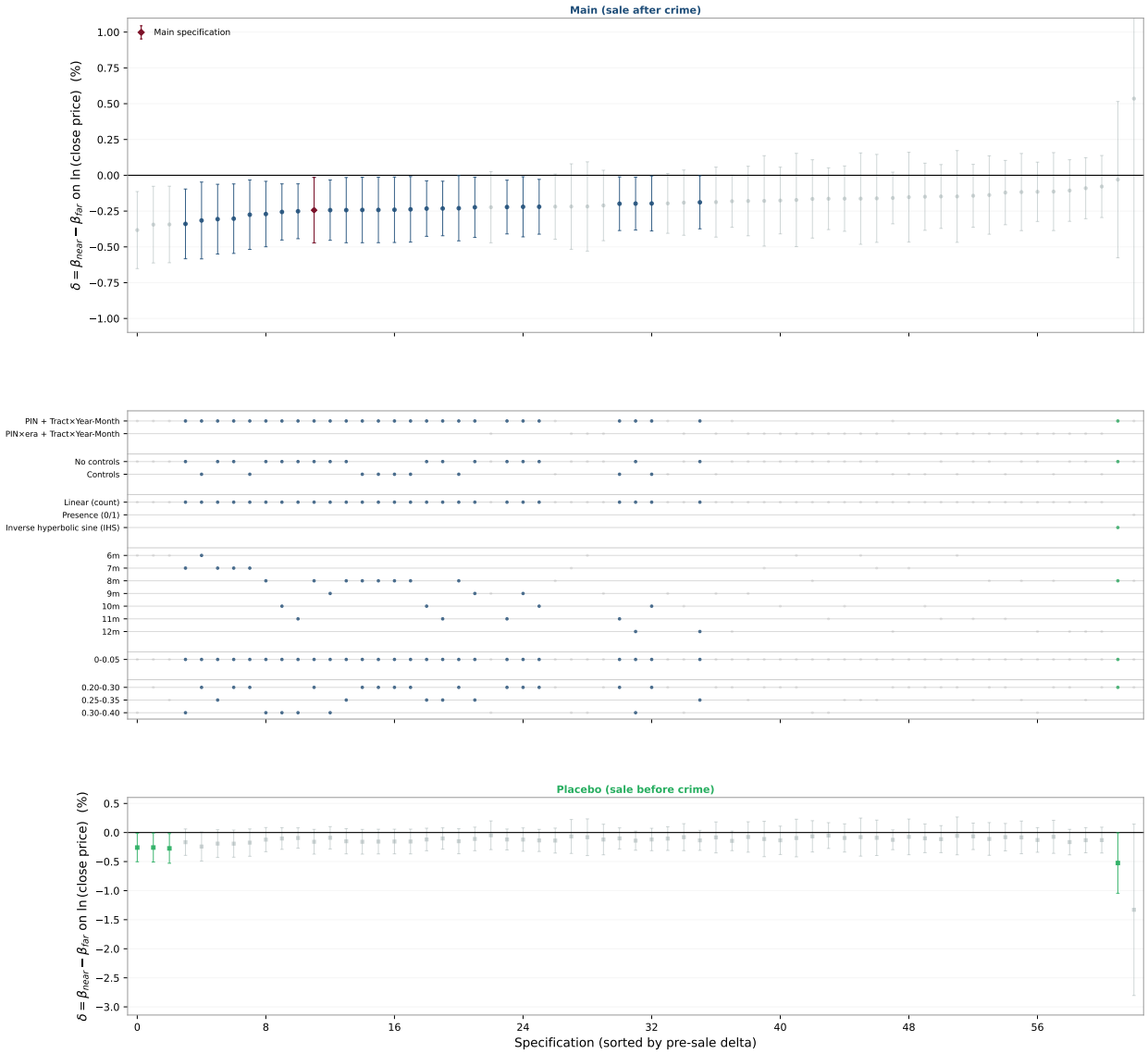


Figure A3: Burglary: Sale-Price Discount Across Specifications

Notes. As in Figure 3, for burglary. Main: 26/63 significant at 5%, all in the discount direction; placebo: 4/63 (the linear count at six months under each of the three far rings, and the inverse hyperbolic sine at eight months).

Supplementary Material

See the *Online Appendix*.